



National Action Plan

Regional Assessment Process

Version 2 - March 2026

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Changes From Previous Version

This version contains the following changes and refinements from previous iterations.

New indicators assessed:

- 4.2.3 – Strategic and Nature Positive Land-Use Planning

Updates to existing indicators:

- Updated criteria weighting tables and increased the clarity of each indicator to allow users of the methodology to better assess their regions (19 indicators: 1.1.1-1.1.5, 2.1.1, 3.1.1-3.1.6, 3.2.1-3.2.6, 3.3.1).
- Score descriptions updated for the following indicators:
 - 1.1.1 – Renewable Energy Generation
 - 1.1.2 – Transmission Network
 - 1.1.3 – Distribution Network
 - 1.1.4 – Energy Storage
 - 1.1.5 – Hydrogen Network
 - 2.1.1 – Industry Decarbonisation Plans
- Updated dimensions / elements to be reviewed in the assessment for the following indicators:
 - 1.1.1 – Renewable Energy Generation
 - 1.1.2 – Transmission Network
 - 1.1.3 – Distribution Network
 - 1.1.4 – Energy Storage
 - 1.1.5 – Hydrogen Network

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Reviewers do not imply any endorsement of the content of this report.

Context

The need for a National Action Plan

Australia has the opportunity to become a global leader in clean exports, creating new industries and quality jobs while cutting emissions. But making this happen requires action that is coordinated, place-based, and driven by a clear-eyed understanding of what makes projects succeed.

Australia's clean energy rollout is already well underway, and policies supporting clean exports are growing at federal and state levels. But the scale of the transformation means that investment, infrastructure, and workforce planning need better alignment. Without stronger coordination, we risk slow, fragmented action that leaves communities behind and projects stalled.

This work complements recent policy reviews such as the Climate Change Authority's Sector Pathways Review, which highlights the need for greater policy coordination, faster project approvals, and more regionally tailored transition planning.

The **National Action Plan** (NAP) exists to address the challenges of coordination across sectors by providing a place-based approach to ensure the successful delivery of major projects. Without coordinated action, the risk is that investment, infrastructure, and workforce planning will lag behind the needs of the industries that will drive Australia's future prosperity.

The urgency of this work cannot be overstated. The global economy is rapidly shifting away from fossil fuels, and regions that do not act now risk losing out on new industries, jobs, and investment. By collaborating on the National Action Plan today, we can ensure that Australia moves quickly to secure its place as a global clean energy leader.

This approach is grounded in a set of Critical Success Criteria—five themes, 17 categories, and 52 indicators—developed through consultation with industry, government and community experts to reflect what is essential for the success of major projects in a zero-emissions economy.

At Beyond Zero Emissions, we've developed this framework because of our long-standing focus on identifying the strongest levers for systems change. Our research has helped shape the policies and investment priorities underpinning Australia's renewable energy future. With deep relationships across sectors, we are now leading the development of the National Action Plan to help decision-makers focus on what matters most—and where.

Why This Guide?

Australia can lead the world in clean industries, powered by abundant renewable energy and critical minerals. But delivering a zero-emissions economy requires coordination and strategic investment.

This guide introduces a structured process for assessing how well regions are positioned to deliver clean industry projects, highlighting strengths, gaps, and priorities for support. At its heart is a set of evidence-based indicators that reflect what it takes to deliver successful clean energy and industries.

Whether you are in local government, industry, community, research, or regional development, this process helps identify your region's strengths and challenges and where to focus action. We welcome collaborators to apply the process, share lessons, and shape future iterations of the National Action Plan: Regional Assessment Process.

Introduction to the Regional Assessment Process

How it works

The National Action Plan: Regional Assessment Process was developed with input from industry, government, community and research stakeholders. It is designed to help evaluate the conditions required to deliver clean industry projects across regions, sectors, and levels of government. It is structured around a hierarchy of **Themes**, **Categories**, and **Indicators**, known collectively as the **Critical Success Criteria**.

- **Themes** represent the top-level conditions that underpin success—such as enabling infrastructure, policy alignment, and community support.
- **Categories** are sub-areas within each theme, such as water infrastructure, workforce, or federal policy settings.
- **Indicators** are the specific, measurable criteria used to assess whether each category is 'ready' or needs attention.

There are **five Themes**, broken down into **17 Categories**, and a total of **52 Indicators**. These indicators define what good looks like and support consistent, evidence-based assessments of regional and sectoral preparedness for clean industry development.

This document (Version 2, March 2026) provides an assessment process for **20 Indicators** across six Categories and four Themes. The remainder will be developed and tested with partners through ongoing collaboration.

We are seeking partners with sectoral, regional, and technical expertise to:

- Help complete and refine the remaining Indicators
- Apply the current version to real-world contexts
- Share insights to improve future iterations

The goal is a practical and collaborative approach to assessing regional potential that is grounded in data, informed by experience, and responsive to what is needed on the ground.

How we are benchmarking ambitious emissions reduction

While we are assessing industry decarbonisation progress and government policy against a target of 75% emissions reduction below 2005 levels by 2035,¹⁻³ for the electricity grid we are assessing against the target of near 100% renewable generation by 2035. These benchmarks are aligned with limiting the rise of global temperatures to within 1.5°C of global warming.¹⁻³

In National Electricity Market (NEM) states, such an energy system is represented by the most ambitious scenarios in the AEMO's Integrated System Plan. Our assessments to date have been benchmarked against the 2024 ISP's Green Energy Exports scenario, which has been superseded in the Draft 2026 ISP by the Accelerated Transition scenario. Because Accelerated Transition assumes lower rates of domestic and industrial electrification than Green Energy Exports (GEE), these ISP forecasts should be tested against the on-the-ground plans and requirements of REIP manufacturers to ensure they provide a sufficiently ambitious benchmark.

Since this methodology was first developed, the Federal Government has announced an interim target of reducing emissions by 62–70% below 2005 levels by 2035. Although industrial decarbonisation policy is likely to be calibrated to this target, we retain the 75% reduction as the benchmark for ambitious emissions reduction in these assessments.

The score descriptions and assessment frameworks used to assess the Indicators to date are shared as part of the documentation below along with weighting tables. These tables apply proportional weights to different dimensions within an Indicator, reflecting their relative importance to overall readiness. The final score for each indicator is calculated from the weighting table output as:

Score	Determination
5 – Excellent	Table output ≥ 4.5
4 – Good	$3.5 \leq \text{table output} < 4.5$
3 – Developing	$2.5 \leq \text{table output} < 3.5$
2 – Improving	$1.5 \leq \text{table output} < 2.5$
1 – Poor	Table output < 1.5

Using the tool to develop regional insights

The **Regional Assessment Process** is designed to generate actionable insights at the regional level by highlighting both strengths and gaps in readiness for clean industry development. By applying the Critical Success Criteria to a specific location, users can build a clearer picture of where investment, policy alignment, or infrastructure upgrades are most needed.

Each indicator provides a structured way to examine conditions on the ground—such as whether the local electricity network can support clean industry growth, or whether regional workforce plans align with net zero goals. When these assessments are viewed collectively, they offer a powerful, evidence-based snapshot of regional preparedness.

These insights can be used to:

- **Prioritise action** where readiness is lowest and support is most urgently needed
- **Inform infrastructure planning, and regional strategies**
- **Support collaboration** by creating a shared understanding among governments, businesses, communities and investors
- **Track progress** over time by repeating the assessment to monitor change and improvement

The process is designed to be adaptable, allowing regions to draw on available data, local expertise, and stakeholder input to shape meaningful, place-based recommendations.

Table 1. Critical Success Criteria Themes, Categories and Indicators

 Enabling Infrastructure	 Industry Capability & Capacity	 Policy & Governance	 Social Acceptance	 Financing the Transition
Energy System Renewable energy generation Transmission network Distribution network Energy storage Hydrogen network Energy system orchestration	Industry Decarbonisation Industry decarbonisation plans Workforce Capacity to Deliver Existing workforce capacity Reskilling and upskilling programs	Federal Policy Emissions reduction commitment Skills and training Approvals processes Investment framework Research and development Government procurement support	Community Benefits & Impacts Strategic and nature-positive land-use planning First Nations community benefit sharing Wider community benefit sharing	Investment Public finance commitment Attracting private investment Risk & Resilience Risk and resilience
Transport System Port Rail Road Transport networks	Innovation & Knowledge Development Innovation Commercialisation Knowledge generation and sharing	State Policy Emissions reduction commitment Skills and training Approvals processes Investment framework Research and development Government procurement support	Engagement & Participation of Communities Effective First Nations engagement Effective community engagement	
Water System & Waste Management Water supply Wastewater management Solid waste & resource recovery	Collaboration & Industrial Ecosystems Industrial symbiosis & material exchange Coordinated infrastructure plan for common user infrastructure Contribution to Australian supply chain resilience International linkages / international coordination	Local Government Policy LGA decarbonisation plans Regional climate adaptation/resilience plan		
Social Infrastructure Housing Health Education		Governance Regional coordination Regional representation of key agencies Policy alignment and coordination		
Communications Infrastructure Digital infrastructure				

 Evaluated as part of preliminary assessments
 Criteria yet to be investigated

This Edition

The **National Action Plan: Regional Assessment Process (Version 2)** is designed to be iterative. This second release (March 2026) outlines the updated assessment process for 20 Criteria across six Categories and four Themes. Before using this process to generate insights for a region, it is important to check that you are referring to the most recent version of an indicator's methodology, as listed in Appendix A.

This body of work is a collaborative effort, including the design of the assessment process itself, which determines what "good" looks like for each scoring band. We welcome feedback and input to ensure the Criteria remains relevant and accurate.

Please note: Individual criteria alone do not generate meaningful insights; it is the collective analysis to build a macro view across different sectors that adds value. The results are intended to be shareable, and we encourage stakeholders to distribute them to others interested in this project.

Regional Assessment Process

1. Enabling Infrastructure

Definition: *Enabling infrastructure refers to the full suite of infrastructure required to facilitate the decarbonisation of industry, including key energy, transport, waste and water infrastructure dependencies of the transition, and the social infrastructure required to accommodate the influx of workers and their families. The first phase of the National Action Plan covers enabling infrastructure associated with the Energy System only.*

Measure: Readiness of physical infrastructure required to facilitate the energy transition and the development of clean industry.

Why is meeting this criteria critical? Enabling infrastructure for the energy system is essential for integrating renewable energy, electrifying industrial processes, and supporting green fuels and feedstocks like hydrogen. These are crucial for reducing emissions, enabling clean commodity manufacturing and decarbonising cleantech manufacturing.

1.1 Energy System

The energy system assessments examine whether current and planned projects are sufficient to meet the needs of a regional grid approaching 100% renewable generation by 2035, while still supporting substantial industrial electrification and the growth of new green industries.

To date, these assessments have used the [2024 Integrated System Plan](#) (ISP) Green Energy Exports scenario published by the Australian Energy Market Operator as the benchmark for such a system. The methodology set out below is therefore defined with reference to this scenario.

In the [Draft 2026 ISP](#), Green Energy Exports has been superseded by the Accelerated Transition scenario. Because Accelerated Transition assumes lower levels of domestic and industrial electrification than Green Energy Exports, ISP demand forecasts should be tested against the on-the-ground plans and requirements of REIP manufacturers before being relied upon as a sufficiently ambitious benchmark.

1.1.1 Renewable Energy Generation

Purpose of assessment: to evaluate the readiness of renewable energy (RE) generation to meet the demand from clean industry and green exports by 2035.

The assessment compares the regional renewable energy project pipeline against a grid approaching zero operational emissions, powered by nearly 100% renewable energy by 2035. This is generally consistent with the Green Energy Exports scenario in AEMO's Integrated System Plan.

Assessment framework

The assessment framework evaluates the energy system based on the following key dimensions:

1. **Current Renewable Energy Share** – Percentage of electricity generation from renewables over the past 12 months.
2. **Renewable Energy Pipeline Status:**
 - a. **Overall Pipeline** – MW capacity of RE projects in operation, approved, or in design/planning relative to required capacity to meet forecast demand under a clean industry/green export scenario by 2035.
 - b. **Approved Pipeline** – Percentage of required renewable energy projects (measured by capacity) that have secured planning approval.
 - c. **Pipeline Progression** – Percentage of required renewable energy capacity that is in construction or operating.
3. **Timing & Delivery Risk** – Feasibility of delivering the required renewable energy capacity on schedule, based on an assessment of current project approvals, lead times, and delivery timelines, as well as major bottlenecks and historical delays.

Data and evidence sources

Data sources include industry reports, regulatory filings, and independent energy system assessments to ensure a comprehensive evaluation:

Renewable Energy Build-Out Trajectory

- [OpenNEM](#) for real-time renewable generation data (Renewable energy current generation status)
- AEMO ISP [Generation & Storage Outlook](#)
- [Clean Energy Regulator \(CER\) data](#)
- [RenewMap](#) for project pipeline progress data to track against the ISP data (a proprietary catalogue of Australia's energy projects)
- BRC-A's State of the Australian Corporate PPA Market [Report](#)
- AEMO's [NEM Generation Information](#) datasets

Regional Electricity Demand

- Distribution Network Service Provider (DNSP) and Transmission Network Service Provider (TNSP) data for current electricity demand
- Forecasted energy demand under AEMO ISP Step Change and Green Energy Exports scenarios
- Regional energy demand modelling, e.g. BZE's REIP Briefing Papers

Scoring process

1. **Assess the current share of renewable energy generation** using OpenNEM for the state the region is located in over the past 12 months.

2. **Compare operational projects and current pipeline** against a scenario aligned to the target of near 100% renewable generation by 2035 (Green Energy Exports)
 - a. **Define geographic scope:** Determine which ISP Renewable Energy Zones (REZs) are assumed to service the Precinct.
 - b. **Analyse project pipeline:** Use RenewMap to quantify solar and wind projects (measured in MW) within the defined geographic scope across development stages: operational, under construction, approved, in planning, and pre-planning/design. Include relevant non-REZ projects within this broader geographic region.
 - c. **Benchmark progress:** Compare total pipeline capacity, approved projects, and projects under construction or in operation against the 2035–36 requirements in the Green Energy Exports scenario. Use these percentages to score the relevant dimensions in the weighting table below.
 - d. **Sense-check:** Estimate electricity demand from large energy users based on publicly available reports to sense-check alignment with the Green Energy Exports scenario.
 - e. **Optional:** Compare the project pipeline against Step Change requirements to quantify the ambition gap between current development and required planning.
3. **Assess sources of delay:** Review the historical incidence of project delays using AEMO's Generation Information database where possible. Identify systematic constraints, including development approval timelines, federal environmental approvals, community opposition, financing delays, government opposition, grid connection issues, workforce shortages, supply chain disruptions, and regulatory changes.
4. **Score Calculation:** Assign a weighted score based on the overall assessment across the evaluation dimensions.
5. **Stakeholder Insights and Industry Reports:** Validate findings through industry expert consultations and energy market assessments.
6. **Review:** Where relevant, integrate stakeholder insights to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Current Renewable Energy Share	0.1	<10% of state's current electricity consumption	≥10% to <30% of state's current electricity consumption	≥30% to <55% of state's current electricity consumption	≥55 to <90% of state's current electricity consumption	≥90% of state's current electricity consumption
Overall Pipeline	0.3	<25% of 2035 green energy exports requirements	≥25% to <50% of 2035 green energy exports requirements	≥50 % to <75% of 2035 green energy exports requirements	≥75% to <100% of 2035 green energy exports requirements	≥100% of 2035 green energy exports requirements

Approved Pipeline	0.2	<15% of 2035 green energy exports requirements	≥15% to <40% of 2035 green energy exports requirements	≥40% to <60% of 2035 green energy exports requirements	≥60 to <85% of 2035 green energy exports requirements	≥85% of 2035 green energy exports requirements
Pipeline Progression	0.25	<10% of 2035 green energy exports requirements	≥10% to <25% of 2035 green energy exports requirements	≥25% to <40% of 2035 green energy exports requirements	≥40% to <50% of 2035 green energy exports requirements	≥50% of 2035 green energy exports requirements
Timing & Delivery Risk	0.15	≥4 categories flagged	3 categories flagged	2 categories flagged	1 category flagged	No categories flagged

Score descriptions

Score	Description	Key Characteristics
1: Not Ready	Poor progress towards 2035 Renewable Energy Zone (REZ) requirements.	- Less than 25% of required capacity is in development - 15% or less of required capacity is approved - 10% or less in construction or operating - Numerous systemic issues contributing to delays
2: Basic Readiness	Improving progress towards 2035 REZ requirements.	- Between 25% and 50% of required capacity is in development - 15-40% of required capacity approved - 10-25% of required capacity in construction or operating - Several systemic factors contributing to delays
3: Developing Readiness	Moderate progress towards 2035 REZ requirements.	- Between 50% and 75% of required capacity is in development - 40-60% of required capacity approved - 25-40% in construction or operating. - A couple of systemic factors contributing to delays
4: Strong Readiness	Strong progress towards 2035 REZ requirements.	- Between 75% and 100% of required capacity is in development - 60-85% of required capacity is approved - 40-50% in construction or operating - Systemic issue contributing to delays

5: Exemplary Readiness	Exemplary progress towards 2035 REZ requirements.	- More than 100% of required capacity is in development - Over 85% of required capacity approved - Over 50% in construction or operating. - Little risk of delay
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1.1.2 Transmission Network

Purpose of assessment: to evaluate the readiness of the high voltage transmission network to connect and deliver the RE generation required to meet demand from clean industry and green exports by 2035.

The assessment considers whether current and planned transmission projects are sufficient to support a grid powered by nearly 100% renewable energy by 2035, aligned with the Green Energy Exports scenario in AEMO's 2024 ISP.

Assessment framework

Transmission network readiness is assessed based on the following key dimensions:

1. **Transmission Network Service Provider (TNSP) Forecasting** – Is the TNSP using scenario-based modelling for future annual electricity consumption and peak demand, and is this in line with AEMO's forecasts?
2. **Optimal Development Path Progression** – Are the *committed*, *anticipated* and *actionable* ISP optimal development path (ODP) projects relevant to the industrial precinct tracking to meet their Green Energy Exports optimal in-service dates?
3. **Future Project Progression** – Are relevant projects designated as *future* in the ODP and *actionable* in the Green Energy Exports least-cost development path progressing according to their optimal in-service date?
4. **REZ Rollout** – Is Renewable Energy Zones development proceeding effectively and transparently, and are REZ-delivery plans in-line with the Green Energy Exports?

Data and evidence sources

Data sources include industry reports, regulatory filings, and independent energy system assessments to ensure a comprehensive evaluation. Key data collection activities include:

- Identify transmission projects linked to region in AEMO ISP [Appendix 5 Network Investments](#) for both:
 - The *Optimal Development Path* – i.e. the final development path suggested by AEMO after marginalising over all scenarios
 - Green Energy Exports *Least-Cost Development Path*
- Publicly available project information from transmission networks, e.g. EnergyCo for the [Hunter Transmission Project](#)
- TNSP Transmission Annual Planning Reports
- REZ planning updates
- AEMO's [Electricity Forecasting Data Portal](#)

Scoring process

- 1. Forecasting Assessment:** benchmark TNSP's 2035 forecast annual delivery electricity and peak demand for the state housing the REIP against AEMO's Green Energy Exports scenario.
- 2. Preliminary Analysis:** Review existing transmission network data from sources noted above.
- 3. Identify Transmission Network Capacity Status:**
 - a. Size of REZ(s) powering the location
 - b. Identify transmission linked projects to the region in AEMO ISP Appendix 5 Network Investments. Project descriptions should identify what areas the project will service.
 - c. Analyse projects for existing capacity, additional capacity to be added, expected costs, delivery date for both Step Change and Green Energy Exports and AEMO's status
 - d. Review transmission project status against relevant state planning department (such as NSW planning/EnergyCo)
- 4. Major Transmission Project Assessments:**
 - a. Assess completion trajectory of each relevant project (1–5), with reference to anticipated lead time in AEMO ISP Appendix 5 Network Investments and typical project timeline uncertainties in AEMO's Electricity Network Options Report and Transmission Cost Database. Where possible, compare these to real, recently completed transmission projects. Scores are assigned based on:
 - i. **5** – Completed or near certainty of completion ahead of optimal ISP timing
 - ii. **4** – Likely completion by optimal ISP timing
 - iii. **3** – Likely completion moderately behind optimal ISP timing (up to 1 year)
 - iv. **2** – Likely completion well behind optimal ISP timing (1-2 years)
 - v. **1** – Completion unlikely or significantly behind optimal ISP timing (>2 years)
 - b. Aggregate scores, weighted by additional transmission capacity.
 - c. Sense-check transmission investment against estimated regional demand
- 5. REZ Assessment**
 - a. Review state's official REZ plan and score each REZ's development status according to:
 - i. **5** – REZ officially declared with major transmission upgrades approved/under construction and connection rights granted
 - ii. **4** – REZ officially declared, with connection rights or approvals granted
 - iii. **3** – REZ officially declared, with progress towards awarding connection rights and approvals
 - iv. **2** – Progress towards declaration evident (e.g. assessments and meaningful community engagement underway)

- v. **1** – Little to no progress on REZ development
- b. Scale score for each AEMO REZ by comparing state's anticipated REZ generation with 2035 Green Energy Exports requirements.
- c. Determine final indicator score by averaging AEMO REZ scores, weighted by 2035 Green Energy Exports generation capacity.
6. **Score Calculation:** Assign a weighted score based on the overall assessment across the evaluation dimensions.
7. **Stakeholder Insights and Industry Reports:** Validate findings through transmission industry expert consultations and energy market assessments.
8. **Review:** Where relevant integrate stakeholder insights to finalise the assigned score.

Weighting table

Dimension	Sub-dimension	Weight	Score				
			1	2	3	4	5
TNSP Forecasting	Peak load growth	0.1	<25% of state's 2035 green energy exports peak load	≥25% to <50% of state's 2035 green energy exports peak load	≥50% to <75% of state's 2035 green energy exports peak load	≥75% to <100 of state's 2035 green energy exports peak load	≥100% of state's 2035 green energy exports peak load
	Annual delivered electricity growth	0.1	<25% of state's 2035 green energy exports annual consumption	≥25% to <50% of state's 2035 green energy exports annual consumption	≥50% to <75% of state's 2035 green energy exports annual consumption	≥75% to <100 of state's 2035 green energy exports annual consumption	≥100% of state's 2035 green energy exports annual consumption
Optimal Development Path Progression		0.4	Calculated as average over scores for each project (weighted by additional transfer capacity into REIP)				
Future Project Progression		0.2	Calculated as average over scores for each project (weighted by additional transfer capacity into REIP)				
REZ Rollout		0.2	Score each REZ by progress, then average (weighted by anticipated generation capacity in REZ)				

Scoring description

Score	Description	Key Characteristics
1: Limited Readiness	Minimal Transmission Network Service Provider (TNSP) planning for industrial electrification	- Required AEMO Integrated System Plan (ISP) Optimal Development Path (ODP) projects well behind schedule - Limited Renewable Energy Zone (REZ) development
2: Basic Readiness	Planning for partial decarbonisation of existing industry	- Some ODP actionable projects on schedule - Preparatory REZ work underway but no declarations.
3: Developing Readiness	Planning for full decarbonisation of existing industry	- All ODP actionable projects on schedule - No progress on additional transmission projects required for Green Energy Exports development path - Moderate REZ progress with some declarations
4: Strong Readiness	Planning for full decarbonisation & modest new green industries	- All actionable ISP projects on schedule - Some progress on additional Green Energy Exports projects - All relevant REZs are declared with connection rights and transmission buildout on track, but short of Green Energy Exports requirements
5: Exemplary Readiness	Planning for full decarbonisation & new green export industries	- All actionable Green Energy Exports transmission projects on schedule - REZ development advanced to meet or exceed Green Energy Exports requirements.

1.1.3 Distribution Network

Purpose of assessment: To evaluate the readiness of the distribution network (the lower voltage power network) to manage two-way power flows and rising electrification from residential and commercial consumer energy resources (CER) through to 2035.

This assessment considers whether distribution network planning is adequate to support CER uptake and residential and industrial electrification consistent with a grid powered by nearly 100% renewable energy by 2035, as represented by the Green Energy Exports scenario in AEMO's 2024 ISP.

Note: AEMO's data does not distinguish between network type (transmission versus distribution), meaning a fully quantitative analysis is not currently possible. Industrial electrification planning is assessed on a qualitative basis as outlined below.

Assessment framework

The assessment framework evaluates the low-voltage distribution network based on the following key dimensions:

1. **CER Uptake Forecasts** – The extent to which DNSPs' forecasts of rooftop solar, behind-the-meter batteries and electric vehicle uptake align with the 2035 Green Energy Exports scenario.
2. **CER Programs and Initiatives** – The strength of programs designed to support projected CER uptake and better network utilisation, such as dynamic household connections, appropriately priced community batteries, or favourable electric vehicle tariffs.
3. **Hosting Capacity Data** – The availability, quality, granularity and completeness of network hosting capacity maps or databases for medium and low voltage feeders.
4. **Network Demand Forecasting** – The availability, time horizon, transparency and spatial granularity of DNSP network demand forecasts.
5. **Substation and Bulk Supply Point Forecasting** – The availability, time horizon and transparency of scenario-based forecasts for bulk supply points and zone substations.

Data and evidence sources

To ensure a comprehensive evaluation, data is collected from multiple credible sources:

- **Distribution Network Operators & Infrastructure Entities:**
 - Distribution Annual Planning Reports from relevant distribution network service providers (DNSPs), e.g. Ergon Energy in regional Queensland.
 - Network development plans, investment roadmaps, and regulatory filings, e.g. RIT-D investment submissions
 - AEMO's [Electricity Forecasting Data Portal](#)
- **Government and Regulatory Reports:**

- Government policy documents, climate roadmaps, and regulatory filings on electrification and grid modernisation.
- **Industry and Expert Assessments:**
 - Independent industry reports on network capacity constraints and expansion plans.
 - Expert analyses on grid integration challenges and opportunities.
- **Stakeholder Engagement:**
 - Conduct consultations with distribution service providers, energy market regulators, and major energy users.
 - Engage with industry stakeholders and businesses relying on electrification for operational transition.

Scoring process

1. **CER Uptake Forecasts:** Analyse CER uptake forecasts used in most recent distribution annual planning report(s) and compare with 2035 Green Energy Exports values.
2. **CER Programs:** Catalogue DNSP's CER-related programs and tariffs and assess the extent to which they align with DNSP and AEMO forecasts.
3. **Hosting Capacity Database/Map:** Determine whether such a map exists and, if so, the frequency of updates, granularity, completeness and quality of data.
4. **Study Network-Wide and Network Node Forecasts:** Examine DNSP forecasts across the network, bulk supply points and zone substations. Determine whether long-horizon, scenario-based forecasts are publicly available with end-use categorisation and regional granularity.
5. **Score Calculation:** Assign a weighted score based on the overall assessment across the evaluation dimensions.
6. **Stakeholder Insights and Industry Reports:** Validate findings through industry expert consultations and energy market assessments.
7. **Review:** where relevant integrate stakeholder insights to finalise the assigned score.

Note: the role of DNSPs in the transition is likely to change significantly in light of the inclusion of distribution network augmentation options and in AEMO's ISP from 2026 onwards. Similarly, demand-side opportunities will be included in subsequent iterations of the ISP. These will require this assessment framework to be updated. This will require greater complementarity between AEMO and DNSP data, which will enable future assessments to be more quantitative.

Weighting table

Dimension	Sub-dimension	Weight	Score				
			1	2	3	4	5
CER Uptake Forecasts	Solar PV uptake	0.067	<25% of 2035 green energy exports forecast	≥25% to <50% of 2035 green energy exports forecast	≥50 % to <75% of 2035 green energy exports forecast	≥75% to <100 of 2035 green energy exports forecast	≥100% of 2035 green energy exports forecast
	EV uptake	0.067	<25% of 2035 green energy exports forecast	≥25% to <50% of 2035 green energy exports forecast	≥50 % to <75% of 2035 green energy exports forecast	≥75% to <100 of 2035 green energy exports forecast	≥100% of 2035 green energy exports forecast
	Battery uptake	0.067	<25% of 2035 green energy exports forecast	≥25% to <50% of 2035 green energy exports forecast	≥50 % to <75% of 2035 green energy exports forecast	≥75% to <100 of 2035 green energy exports forecast	≥100% of 2035 green energy exports forecast
CER Programs and Initiatives		0.2	Network policies not aligned with enabling CER	Partial network policy alignment with enabling CER	Moderate network policy alignment with enabling CER	Good network policy alignment with enabling CER	Full network policy alignment with enabling CER
Hosting Capacity Data		0.2	Feeder hosting capacity map/database not provided or limited in scope	MV feeder hosting capacity database provided with 2 of: - CER hosting capacity - Utilisation - End-use categorisation - Granularity down to local distribution transformers	MV feeder hosting capacity database provided with 3 of: - CER hosting capacity - Utilisation - End-use categorisation - Granularity down to local distribution transformers	MV feeder hosting capacity database provided with all of: - CER hosting capacity - Utilisation - End-use categorisation - Granularity down to local distribution transformers with some gaps	MV feeder hosting capacity database provided with all of: - CER hosting capacity - Utilisation - End-use categorisation - Granularity down to local distribution transformers with no gaps
Network Demand Forecasting		0.2	Network-wide forecasts available only	Network-wide scenario-based forecasts publicly available with no categorisation	Scenario-based forecasts available with either basic end-use/customer -type categorisation or regional breakdowns	Regional scenario-based forecasts available with basic end-use categorisation	Regional scenario-based forecasts available with full end-use categorisation
Substation and Bulk Supply Point Forecasting		0.2	Single, 5-year forecasts available	Forecasts available with one of: - 10-year period horizon - different scenarios - basic end-use categorisation	Forecasts available with two of: - 10-year period horizon - different scenarios - basic end-use categorisation	10-year scenario-based forecasts available with basic end-use categorisation	Long-term, scenario-based forecasts available with full end-use categorisation

Scoring descriptions

Score	Description	Key Characteristics
1: Limited Readiness	No plans for electrification including at the household level	- No plans for household Consumer Energy Resources (CER) support - No plans to support industrial electrification
2: Basic Readiness	Limited electrification planning	- Planning for modest consumer energy resources (CER) uptake - No plans to support industrial electrification - Limited forecasting and network utilisation transparency
3: Developing Readiness	Moderate electrification planning	- Planning for high uptake of CER - No plans to support industrial electrification - Improving forecasting and network utilisation transparency
4: Strong Readiness	Strong electrification planning	- Planning for high uptake of CER and a modest level of industry electrification. - Strong forecasting and network utilisation transparency.
5: Exemplary Readiness	Exemplary electrification planning	- Planning for high CER uptake and industrial electrification. - Comprehensive forecasting and network utilisation transparency.

1.1.4 Energy Storage

Purpose of assessment: To evaluate the readiness of regional energy storage systems to support a 100% renewable energy supply capable of meeting demand from clean industry and green exports by 2035.

The assessment compares the regional energy storage project pipeline with Australia approaching a zero emissions grid powered by 100% renewable energy by 2035 – generally consistent with AEMO's ISP Green Energy Exports scenario.

Assessment framework

The assessment framework evaluates the energy storage system based on the following key dimensions:

1. Renewable Energy Storage Pipeline Status:

- a. **Overall Pipeline** – Cumulative size* of storage projects in development or further progressed relative to required capacity to meet forecast demand under a clean industry/green export scenario by 2035.
- b. **Approved Pipeline** – Percentage of required storage projects that have secured planning approval.
- c. **Pipeline Progression** – Percentage of required storage projects that are in construction or operation.

2. Timing & Delivery Risk – Feasibility of delivering the required renewable storage on schedule, based on an assessment of current project approvals, lead times, and delivery timelines, as well as major bottlenecks and historical delays.

* These pipeline metrics are evaluated twice – once using the charging/discharging capacity (i.e. power) of storage projects, and again using their depth (storage energy). This means that both the instantaneous discharging rate and duration are being assessed.

Data and evidence sources

Data sources include industry reports, regulatory filings, and independent energy system assessments to ensure a comprehensive evaluation. Key data collection activities include:

Renewable Energy Storage Status

- OpenNEM for real-time renewable storage data
- Australian Energy Market Operator (AEMO) Integrated System Plan (ISP) for projected storage capacity
- RenewMap for project tracking of energy storage projects (operational, approved, in planning and announced)
- Aggregate data on utility storage energy projects at state and regional levels

- State databases for large-scale infrastructure projects like Pumped Hydro (e.g. Queensland Coordinated Project database)

Renewable Energy Build-Out Trajectory

- AEMO ISP Step Change Scenario for projected storage capacity
- AEMO ISP Green Energy Exports scenario targets for projected storage capacity
- Regional project pipeline data from RenewMap for storage infrastructure.

Regional Electricity Demand

- Energy demand estimates used for indicator 1.1.1 for consistency
- Assessment of industrial and grid-scale storage needs in relation to overall energy system requirements.

Industry & Expert Reports

- Independent energy storage assessments, e.g. Nexa Advisory.
- Project tracking database for energy storage developments.

Note: Storage buildout is reported in the ISP in terms of discharge capacity and energy at the sub-regional level, rather than by REZ. The assessor should identify the main ISP sub-regions relevant to the REIP under assessment as well as those which host the key REZs.

Scoring process

1. **Compare operational projects and current pipeline** against a scenario aligned to the target of near 100% renewable generation by 2035 (Green Energy Exports)
 - a. **Define geographic scope:** Determine which ISP sub-regions are assumed to service the Precinct. If possible, this should be consistent with the renewable energy zones used in the evaluation of 1.1.1, but storage may have to be assessed at the state-level if sub-regions are not appropriate.
 - b. **Analyse project pipeline:** Use RenewMap to quantify storage projects (measured in MW and MWh) within the defined geographic scope across development stages: operational, under construction, approved, in planning, and pre-planning/design.
 - c. **Benchmark progress:** Compare total pipeline capacity, approved projects, and projects under construction or in operation against the 2035–36 requirements in the Green Energy Exports scenario. Use these percentages to score the relevant dimensions in the weighting table below.
 - d. **Optional:** Compare the project pipeline against Step Change requirements to quantify the ambition gap between current development and required planning.
2. **Assess Sources of Delay:** Assess quantitative history of storage project delays in AEMO's Generation Information or other databases. Flag historical issues with development approval timelines, EPBC approval timelines, community opposition, financing delays, government

opposition, grid connection issues, workforce shortages, supply chain disruptions and changes in regulation which impact these items.

3. **Score Calculation:** Assign a weighted score based on the overall assessment across the evaluation dimensions.
4. **Stakeholder Insights and Industry Reports:** Validate findings through industry expert consultations and energy market assessments.
5. **Review:** Where relevant, integrate stakeholder insights to finalise the assigned score.

Weighting table

Dimension	Sub-dimension	Weight	Score				
			1	2	3	4	5
Overall Pipeline	Capacity (MW)	0.15	<25% of 2035 Green Energy Exports requirements	≥25% to <50% of 2035 Green Energy Exports requirements	≥50% to <75% of 2035 Green Energy Exports requirements	≥75% to <100 of 2035 Green Energy Exports requirements	≥100% of 2035 Green Energy Exports requirements
	Depth (MWh)	0.15	<25% of 2035 Green Energy Exports requirements	≥25% to <50% of 2035 Green Energy Exports requirements	≥50% to <75% of 2035 Green Energy Exports requirements	≥75% to <100 of 2035 Green Energy Exports requirements	≥100% of 2035 Green Energy Exports requirements
Pipeline Progression	Capacity (MW)	0.125	<10% of 2035 Green Energy Exports requirements	≥10% to <25% of 2035 Green Energy Exports requirements	≥25% to <40% of 2035 Green Energy Exports requirements	≥40% to <50% of 2035 Green Energy Exports requirements	≥50% of 2035 Green Energy Exports requirements
	Depth (MWh)	0.125	<10% of 2035 Green Energy Exports requirements	≥10% to <25% of 2035 Green Energy Exports requirements	≥25% to <40% of 2035 Green Energy Exports requirements	≥40% to <50% of 2035 Green Energy Exports requirements	≥50% of 2035 Green Energy Exports requirements
Approved Pipeline	Capacity (MW)	0.125	<10% of 2035 Green Energy Exports requirements	≥10% to <40% of 2035 Green Energy Exports requirements	≥40% to <60% of 2035 Green Energy Exports requirements	≥60 to <90% of 2035 Green Energy Exports requirements	≥90% of 2035 Green Energy Exports requirements
	Depth (MWh)	0.125	<10% of 2035 Green Energy Exports requirements	≥10% to <40% of 2035 Green Energy Exports requirements	≥40% to <60% of 2035 Green Energy Exports requirements	≥60 to <90% of 2035 Green Energy Exports requirements	≥90% of 2035 Green Energy Exports requirements
Timing & Delivery Risk		0.2	4+ factors flagged	3 factors flagged	2 factors flagged	1 factor flagged	No factors flagged

Score descriptions

Score	Description	Key Characteristics
1: Limited Readiness	Poor progress towards 2035 storage requirements.	- Less than 25% of required capacity/depth is in development - 15% or less approved - 10% or less in construction or operating - Numerous systemic issues contributing to delays
2: Basic Readiness	Improving progress towards 2035 storage requirements.	- Between 25% and 50% of required capacity/depth is in development - 15–40% approved - 10–25% in construction or operating. - Several systemic factors contributing to delays
3: Developing Readiness	Moderate progress towards 2035 storage requirements.	- Between 50% and 75% of required capacity/depth is in development - 40–60% approved - 25–40% in construction or operating - A couple of systemic factors contributing to delays
4: Strong Readiness	Strong progress towards 2035 storage requirements.	- Between 75% and 100% of required capacity/depth is in development - 60–85% approved - 40–50% in construction or operating - Systemic issue contributing to delays
5: Exemplary Readiness	Exemplary progress towards 2035 storage requirements.	- Over 100% of required capacity/depth is in development - More than 85% approved - More than 50% in construction or operating - Little risk of delay

1.1.5 Hydrogen Network

Purpose of assessment: To evaluate the readiness of the green hydrogen supply and network to meet demand from clean industry and green exports by 2035.

The assessment considers the adequacy of the green hydrogen supply and network pipeline, project approval status, and alignment with the Green Energy Exports scenario in AEMO's Integrated System Plan. These factors should be cross-checked against the specific requirements of existing industry and prospective green manufacturing in the REIP.

Assessment framework

The assessment framework evaluates the hydrogen energy supply and network based on the following key dimensions:

1. Hydrogen Pipeline Status:

- a. **Overall Pipeline** – Production capacity (in kt per year) of hydrogen projects in development or beyond relative to required capacity to meet forecast demand under a clean industry/green export scenario by 2035.
- b. **Approved Pipeline** – Percentage of required hydrogen production capacity that has secured planning approval.
- c. **Pipeline Progression** – Percentage of required hydrogen production capacity that is in construction or operation.

2. Timing & Delivery Risk – Feasibility of delivering the required hydrogen production capacity on schedule, based on an assessment of the status of individual projects, including approvals, lead times, and delivery timelines, as well as major bottlenecks and historical delays.

Data and evidence sources

To ensure a comprehensive evaluation, data is collected from multiple credible sources:

• Hydrogen Energy Supply and Network Pipeline Status:

- Assessment of government and corporate hydrogen project plans and announcements. (e.g. [Australian Hydrogen Projects Dataset](#), [Australia's Hydrogen Opportunities Tool](#))
- Capital allocation from government and private sector investments.
- Project tracking databases detailing operational, approved, and planned hydrogen infrastructure, e.g. CSIRO's [HyResource](#) and the IEA's [Hydrogen Production and Infrastructure Projects Database](#).

• Integration with Enabling Infrastructure:

- Transmission Network (Indicator 1.1.2): Evaluation of whether transmission infrastructure will be ready to supply sufficient electricity for hydrogen production.
- Water Supply and Waste Management (Indicators 1.3.1 and 1.3.2): Future assessment of water resources required for hydrogen production and related waste management

systems.

- **Industry & Expert Reports:**

- Independent hydrogen network assessments from energy market analysts and industry stakeholders, e.g. [ACIL Allen's Fuel Price Report](#).
- Reports from regulatory agencies on hydrogen infrastructure policy and development.

Scoring process

1. **Current Hydrogen Energy Supply and Network Capacity:** Analyse the existing operational hydrogen supply in the assessed REIP in relation to industrial and export demand.
2. **Future Hydrogen Readiness:** Compare planned hydrogen infrastructure development against projected demand in Green Energy Exports/ and Step Change scenarios.
3. **Approval and Implementation Status:** Determine the proportion of hydrogen network projects that have received approval and are under construction.
4. **Timing & Delivery Risk:** Examine each project and determine whether it is at risk of delay.
5. **Score Calculation:** Assign a weighted score based on performance across the evaluation dimensions.
6. **Stakeholder Insights and Industry Reports:** Validate findings through industry expert consultations and energy market assessments.
7. **Review:** Where relevant integrate stakeholder insights to finalise the assigned score.

Note: In the 2024 ISP inputs, assumptions and scenarios report, domestic hydrogen production and consumption is reported at the state-level, while demand from green iron and export is reported nationally. This demand is allocated to sub-regions within the actual ISP based on the least-cost outcome for the development path in question.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Overall Pipeline	0.3	<25% of 2035 requirements	≥25% to <50% of 2035 requirements	≥50 % to <75% of 2035 requirements	≥75% to <100 of 2035 requirements	≥100% of 2035 requirements
Approved Pipeline	0.25	<15% of 2035 requirements	≥15% to <40% of 2035 requirements	≥40% to <60% of 2035 requirements	≥60 to <80% of 2035 requirements	≥80% of 2035 requirements
Pipeline Progression	0.25	<10% of 2035 requirements	≥10% to <25% of 2035 requirements	≥25% to <40% of 2035 requirements	≥40% to <50% of 2035 requirements	≥50% of 2035 requirements
Timing & Delivery Risk	0.2	<25% of 2035 requirements not at risk	≥25% to <50% of 2035 requirements not at risk	≥50 % to <75% of 2035 requirements not at risk	≥75% to <100 of 2035 requirements not at risk	≥100% of 2035 requirements not at risk

Scoring descriptions

Score	Description	Key Characteristics
1: Limited Readiness	Poor progress towards 2035 hydrogen production requirements.	- Less than 25% of required production capacity in development - 15% or less approved - 10% or less in construction or operating - More than 75% of required production capacity at risk of delay
2: Basic Readiness	Improving progress towards 2035 hydrogen production requirements	- Between 25% and 50% of required production capacity in development - 15-40% approved - 10-25% in construction or operating - 50-75% of required production capacity at risk of delay
3: Developing Readiness	Moderate progress towards 2035 hydrogen production requirements	- Between 50% and 75% of required production capacity is in development, with 40-60% approved and 25-40% in construction or operating. - 50-25% of required production capacity at risk of delay
4: Strong Readiness	Good progress towards 2035 hydrogen production requirements	- Between 75% and 100% of required production capacity in development - 60-80% approved - 40-50% in construction or operating - Between 0% and 25% of required production capacity at risk of delay
5: Exemplary Readiness	Excellent progress towards 2035 hydrogen production requirements	- Over 100% of required production capacity in development - Over 80% approved - Over 50% in construction or operating

2. Industry Capability & Capacity

Definition: Industry's capability and capacity to implement net-zero solutions

Measure: How well industries can adapt, innovate, and contribute to achieving a sustainable, low-carbon future including access to sufficient resources.

Why is meeting this criteria critical? Australian industries need both the know-how and access to resources including innovation and skilled workforce to capture zero-emissions opportunities for existing and new markets.

2.1 Industry Decarbonisation

2.1.1 Industry Decarbonisation Plans

Purpose of assessment: To evaluate the decarbonisation progress of energy-intensive manufacturing industries within the region. The assessment considers the presence of decarbonisation plans, target ambition, funding, and implementation progress. While coal and gas extraction industries are excluded from the primary assessment, their role is noted separately for contextualisation. New projects that are designed to decarbonise supply chains are assessed according to whether their operations use 100% renewable energy (electricity plus heat) from the start. Where zero/low emissions technology is still in development, evidence of clear steps include target ambition, pilot projects, etc.

Assessment framework

The assessment framework evaluates each **industry stakeholder (facility) within the manufacturing sector** with emission reductions above 25,000 tonnes CO₂-e (Scope 1 and 2) against the existence of a decarbonisation plan and evidence of delivery. Scores for each stakeholder are aggregated and averaged to provide the final score.

1. Emissions Reduction Plans

- a. **Baseline Emissions Reduction Plans:** Does the company have a clear and measurable emissions reduction plan/pathway for the facility?
- b. **Alignment with 1.5°C Targets:** Are the company's targets in line with limiting global warming to 1.5°C (75% reduction in emissions by 2035 below 2005 levels)?
- c. **Supply Chain Emissions Considerations:** Does the plan address supply chain emissions? Note: offsets have not been included at this time due to difficulty in obtaining inclusive data.

2. Action:

- a. **Evidence of Action and Funding:** Are tangible steps being taken towards emissions reduction? Is there evidence of funded actions?

- b. **Energy Management and Demand Response:** Is the company addressing its energy efficiency and its ability to manage its energy more effectively especially for load shaping and ability to offer demand response?
3. **Performance Incentives:** Are internal performance incentives (e.g. executive bonuses) linked to meeting emissions targets?

Data and evidence sources

Data sources include corporate sustainability reports, industry disclosures and expert analyses to ensure a comprehensive evaluation. Key data collection activities include:

- **Company Sustainability Reports** – Corporate disclosures on emissions reduction targets and progress.
- **Public Commitments** – Statements from companies outlining decarbonisation intentions.
- **Regulatory Filings & Compliance Reports** – Industry obligations and regulatory adherence including National Greenhouse and Energy Reporting (NGER) Scheme and the Clean Energy Regulator (CER) Safeguard Mechanism reporting.
- **Government & Independent Industry Studies** – Third-party assessments of industry performance, e.g. Climate Action 100, reports by other ENGOS
- **Media Reports & Corporate Announcements** – News coverage of corporate decarbonisation initiatives.
- **Industry Stakeholder & Expert Consultation Interviews** – Insights from sectoral experts and industry representatives.

Scoring Process

1. **Preliminary Analysis:** Review corporate and regulatory disclosures on decarbonisation initiatives.
2. **Indicator Evaluation:** Assess targets, funding mechanisms, implementation strategies, and accountability measures.
3. **Stakeholder Insights:** Engage with industry representatives and sustainability experts to validate findings.
4. **Score Calculation:** Compile individual company assessments and stakeholder insights to determine an overall industry score.
5. **Review:** If major implementation gaps exist—such as technologies that lock in long-term use of fossil fuels, infrastructure delays, or medium-term reliance on fossil fuels—the final score is adjusted accordingly.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Baseline Emissions Reduction Plans	0.25	No clear decarbonisation plans	Some companies have plans but lack clear targets	Most companies have defined plans with targets	All major companies have fully developed 1.5°C-aligned plans	Industry-wide best practice targets and accountability
Alignment with 1.5°C Targets	0.15	No alignment with 1.5°C	Some alignment, but limited scope	Majority of companies have 1.5°C-aligned targets	All major companies have 1.5°C-aligned targets	Industry-wide adoption of science-based targets
Supply Chain Emissions Consideration	0.1	No supply chain emissions tracking	Some indirect emissions tracked but not addressed	Scope 1 & 2 emissions managed, Scope 3 plans emerging	Fully developed Scope 1, 2 & 3 plans with partial implementation	Comprehensive supply chain emissions reduction measures
Evidence of Action & Funding	0.2	No funding allocated for implementation	Some plans exist, but funding mechanisms unclear	Majority of companies have secured funding	Fully funded decarbonisation plans with initial implementation	Industry-wide funding mechanisms and large-scale action
Energy Management & Demand Response	0.1	No energy efficiency strategies in place	Some energy management improvements planned	Most companies implementing demand response & efficiency	Industry-wide adoption of advanced energy management	Leading global best practice in energy efficiency & flexibility
Performance Incentives	0.1	No internal incentives for emissions reduction	Some sustainability commitments but no incentives	Partial incentive structures linked to sustainability	Full incentive mechanisms tied to emissions reductions	Industry-leading financial incentives embedded in governance

Score description

Score	Description	Key Characteristics
1: Limited Plan / Insufficient Targets	Most companies have plans with no targets and/or no funding.	- Weak or symbolic decarbonisation plans. - No clear emissions reduction targets. - No allocated funding for implementation.
2: Basic Plans / Uncertain Implementation Pathways	Most companies have plans with targets, but only some are aligned to 75% emissions reduction by 2035 and not all commitments are funded	- Some decarbonisation plans exist but lack full alignment with climate targets. - Uncertain funding mechanisms or partial commitments. - Implementation pathways unclear or underdeveloped.
3: Intermediate Plans / Developing Momentum	Most companies have plans with targets and most are aligned to 75% emissions reduction by 2035 and are funded	- Majority of companies have defined targets aligned with climate goals. - Funding secured for decarbonisation efforts. - Initial implementation efforts in place.
4: Comprehensive Plans / Aligned with Regional Goals	All major companies have plans with targets aligned with 75% emissions reduction by 2035, with commitments funded but no internal performance initiatives	- Fully developed and funded decarbonisation plans. - Industry-wide alignment with regional net-zero strategies. - Standardised reporting and compliance mechanisms.
5: Best Practice / Exemplary Industry Leadership	All major companies have plans aligned with 75% emissions reduction targets by 2035, are fully funded and have internal performance initiatives	- Industry-wide accountability for full scope 1, 2 and 3 emissions. - Best practice decarbonisation measures embedded in corporate governance. - Financial incentives tied to emissions reductions and sustainability performance.

3. Policy and Governance

Definition: Government policy and programs at the federal, state or local level support accelerated emissions reductions.

Measure: Government policy or program is being well implemented in regions relevant to the generation, supply or consumption of energy, or the manufacture of a good or commodity.

Why is meeting this criteria critical? Effective and well implemented policies are key drivers of rapid emissions reduction, providing the regulatory framework, incentives and mandates necessary for the energy transition. To be effective, policy and programs should be in place, demonstrate uptake, and deliver on ground impact.

3.1 Federal policy

3.1.1 Emissions Reduction Commitment

Purpose of assessment: To evaluate the adequacy of government policies in delivering emissions reduction commitments and driving momentum towards industry decarbonisation and a clean economy. It focuses on target ambition, policy comprehensiveness, regional support, and alignment with the 1.5°C pathway and net-zero goals.

A 75% reduction in economy-wide emissions by 2035 (below 2005 levels) is used as the core benchmark. However, sector-specific benchmarks are applied where appropriate. For example, in the electricity sector, the assessment is aligned with achieving a zero-emissions grid by 2035, consistent with AEMO's Green Energy scenario.

Assessment framework

Policies are evaluated across six areas:

1. **Target Ambition:** Are the emissions reduction targets aligned with current science-based pathways, such as the 1.5°C target?
2. **Policy Development, Regulation and Implementation:** Does the government have detailed, actionable, and enforceable policies to achieve targets?
3. **Sectoral Strategies:** Are sector-specific plans available, funded and integrated with national decarbonisation objectives?
4. **Regional Support:** Is there sufficient support for regions and sectors heavily impacted by the transition? Examples of such support are the Capacity Investment Scheme investments, and other Federal government programs to support emission reductions at the regional level.

5. **Monitoring and Accountability:** Are there mechanisms in place for tracking, reporting, and adjusting targets based on performance? Are targets inclusive of emission sources?

Data and evidence sources

Sources include government policy documents, independent and industry reports and commentary from advisory bodies and stakeholders, in addition to international analysis from leading climate change authorities.

Government Policy Documents

- Climate action strategies and emissions reduction plans
- Budget allocations for climate-related initiatives
- Legislative documents such as climate acts and policy frameworks

Independent Reports & Expert Analyses

- Climate Action Tracker⁷ and similar international rating agencies
- Reports from advisory bodies like the Climate Change Authority (CCA)
- Research from universities and independent think tanks
- Industry & Sectoral Insights

Reports from peak industry bodies (e.g., Australian Industry Group)

- Sectoral performance reports (e.g., energy, transport, agriculture)
- Interviews with industry stakeholders to identify policy impacts

Community & Regional Feedback

- Stakeholder consultations with affected communities
- Submissions to government consultations on climate policy
- Surveys and case studies of transition-impacted regions

International Comparisons

- Analysis of targets and policy frameworks in comparable economies
- UNFCCC reports and other international climate assessment documents

Scoring Process

1. **Preliminary Analysis:** Review federal government emissions reduction policies and legal commitments.
2. **Indicator Evaluation:** Assess target ambition, enforcement mechanisms, and policy coverage.
3. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
4. **Stakeholder Insights:** Engage with government officials, independent agencies, industry representatives, and climate researchers.

5. **Review:** Where relevant, integrate stakeholder insights to finalise the assigned score and identify areas for improvement.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Target Ambition	0.35	Targets are outdated or unambitious, not aligned with science-based pathways	Baseline targets exist but lack strategic clarity	Mid-range targets with some scientific alignment	Ambitious, science-based targets in place	Targets exceed international benchmarks and are best-in-class
Policy Development, Regulation & Implementation	0.25	Symbolic policies with no enforceability	Policies exist but are fragmented and lack funding	Some enforceable measures, but implementation gaps exist	Comprehensive, enforceable policies with clear implementation pathways	Fully integrated policies with strong legal backing and enforcement
Sectoral Strategies	0.15	No sectoral strategies, national policies ignore sector-specific needs	Some sectoral plans exist but are poorly integrated or underfunded	Sectoral strategies emerging but inconsistent	Well-developed, funded sectoral strategies aligned with national goals	Comprehensive sectoral plans fully integrated with national and global frameworks
Regional Support	0.1	No regional support or investment in transition-affected areas	Limited regional initiatives, funding insufficient to drive transition	Growing regional programs but coverage remains uneven	Strong regional investment programs in place, supporting affected industries	Highly effective, well-funded regional support ensuring just transition
Monitoring & Accountability	0.15	No monitoring or tracking mechanisms in place	Some tracking exists, but no clear accountability framework	Basic reporting mechanisms but lack independent oversight	Clear tracking and accountability mechanisms with enforcement measures	Highly transparent and independently verified emissions tracking and reporting

Scoring description

Score	Description	Key Characteristics
1	Minimal Commitment (Severe Deficiency)	- Targets are outdated or unambitious, with symbolic policies lacking enforcement or measurable outcomes. - Regional support is minimal. - No alignment with international standards.
2	Basic Commitment (Moderate Deficiency)	- Baseline targets exist but lack strategic clarity. - Policies are fragmented, underfunded, and fail to drive significant regional engagement.
3	Moderate Commitment (Developing)	- Mid-range targets with growing policy depth. - Sectoral strategies and regional initiatives are developing but remain inconsistent.
4	Strong Commitment (Effective)	- Ambitious, science-based targets with comprehensive, funded policies. - Clear mechanisms for monitoring, enforcement, and regional support achieved.
5	Exemplary Commitment (Leading Practice)	- Targets exceed international benchmarks with holistic, integrated policies. - Strong enforcement, consistent monitoring, and high-impact regional support is in place.

3.1.2 Skills and Training

Purpose of assessment: To evaluate federal government policies related to skills and workforce development for the zero-emissions transition. The assessment focuses on policy presence, effectiveness and investment with a view to identifying gaps and recommending improvements.

Assessment framework

Policies are evaluated across six areas:

1. **Policy Commitment** – Does the policy explicitly recognise skills development as a key driver of the energy transition? Has thorough analysis of the types of skills needed been undertaken?
2. **Funding and Resources** – Are financial and structural resources sufficient to meet skills development needs? Are programs adequately resourced, including expanding training capacity/workforce to be able to deliver programs
3. **Industry and Education Collaboration** – Does the policy facilitate cooperation between government, industries, and educational institutions?
4. **Workforce Planning and Strategy** – Is there a clear, strategic workforce transition plan that aligns with national and regional energy transition goals?
5. **Implementation and Accessibility** – Are programs effectively delivered and accessible to affected sectors and communities?
6. **Outcomes and Continuous Improvement** – Are there mechanisms to track progress, refine strategies, and adapt to emerging challenges?

Data and evidence sources

Government Policy and Budget Documents

- National and regional workforce development strategies
- Budget allocations for training programs
- Parliamentary reports and legislative initiatives

Industry Reports and Expert Analysis

- Reports from key organisations (e.g. Clean Energy Council, Business Council of Australia, Jobs and Skills Australia)
- Labour market studies on skills shortages and emerging workforce trends
- Think tank research on workforce transition needs

Educational and Training Program Reviews

- Availability and effectiveness of clean energy-related vocational training and university courses
- Industry partnerships with training institutions
- Regional skills hubs and their alignment with workforce needs

Regional and Sector-Specific Workforce Data

- Employment trends in renewable energy and decarbonised industries
- Transition challenges for fossil fuel workers in affected regions
- Industry workforce demand projections

Stakeholder Engagement

- Consultations with government agencies, industry leaders, and education providers
- Input from workforce transition groups and unions
- Feedback from impacted workers and job seekers

Scoring process

1. **Preliminary Analysis:** Review government policies and workforce development strategies.
2. **Indicator Evaluation:** Assess funding, industry collaboration, and accessibility of training programs.
3. **Case Study Analysis:** Examine existing workforce transition initiatives and their effectiveness.
4. **Scoring Aggregation:** Assign a score based on performance across key criteria.
5. **Stakeholder Insights:** Engage with industry leaders, training institutions, and workforce representatives.
6. **Review:** Where relevant, integrate stakeholder insights to finalise the assigned score and identify areas for improvement.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Policy Commitment	0.2	No targeted programs or workforce transition recognition	Some recognition but lacks integration with transition planning	Recognises workforce skilling as important but lacks full coordination	Strong commitment with structured policy integration	Comprehensive, long-term commitment with strategic workforce alignment
Funding & Resources	0.2	No funding allocated for workforce transition	Limited or one-off funding for training programs	Some funding but lacks scalability or sustainability	Well-funded, long-term workforce programs with strong institutional backing	Sustainable funding model ensuring long-term industry and regional workforce transformation
Industry & Education Collaboration	0.15	No collaboration between government, industry, and education providers	Some engagement but lacks depth and coordination	Moderate collaboration with emerging industry partnerships	Strong industry partnerships embedded in policy implementation	Deeply integrated government, industry, and education partnerships with continuous feedback mechanisms
Workforce Planning & Strategy	0.15	No workforce transition plan or strategic direction	Basic workforce initiatives exist but lack clear long-term planning	Recognises workforce needs but implementation gaps remain	Comprehensive workforce planning with sectoral strategies	Best-practice workforce strategy, aligned with national and regional economic transitions
Implementation & Accessibility	0.15	No access to reskilling/upskilling programs for workers	Limited access, with barriers for affected workers and communities	Some training programs exist but not widely available or tailored	Well-structured, inclusive training programs supporting multiple sectors	Fully accessible, diverse workforce transition programs with national coverage
Outcomes & Continuous Improvement	0.15	No tracking or evaluation of program effectiveness	Limited monitoring with no process for policy improvement	Some tracking mechanisms but lacks transparent accountability	Well-developed tracking systems with periodic review and improvements	Leading model for adaptive policy, with iterative improvements based on real-world outcomes

Scoring description

Score	Description	Key Characteristics
1	Minimal support (Severe Deficiency)	- Lacks targeted programs, funding, or industry collaboration for zero-emissions skills. - No integration with workforce transition strategies. - No feedback mechanism to improve outcomes.
2	Basic support (Moderate Deficiency)	- Some initiatives exist but lack depth, coordination, or sufficient funding. - Limited industry partnerships and workforce integration.
3	Moderate support (Developing)	- Recognises workforce skilling as important and commits resources but lacks full-scale coordination, accessibility, or regional application. - Some alignment with transition strategies.
4	Strong support (Effective)	- Well-developed programs with substantial funding, strong industry collaboration, and tailored workforce transition strategies. - Supports multiple sectors effectively.
5	Exemplary support (Leading Practice)	- Comprehensive, holistic, and highly coordinated policies with sustainable funding, measurable outcomes, and a clear roadmap for a just transition. - Good feedback mechanism and process for building on learnings to ensure continuous improvement.

3.1.3 Approvals Processes

Purpose of assessment: To evaluate the ability of federal government approvals processes to provide a structured, transparent, and evidence-based framework to facilitate renewable energy and other major infrastructure projects required to deliver decarbonisation of regional industry.

The assessment focuses on process efficiency, regulatory clarity, interlinkage with state government approvals and environmental integration to support timely, sustainable energy transition.

Assessment framework

Policies will be assessed based on six dimensions:

1. **Process Efficiency:** Are approval timelines predictable, efficient, and supportive of timely project delivery? Is there a 'one stop shop' for project approvals for generation and storage infrastructure projects?
2. **Regulatory Clarity:** Are the processes and requirements well-defined and consistent across jurisdictions?
3. **Environmental Integration:** Do the processes include comprehensive, science-based environmental and emissions criteria? Is the application process comprehensive with a balance between environmental concerns and complexity?
4. **Support for Innovation:** Are emerging zero-emissions technologies accommodated with adaptive pathways?
5. **Regional Adaptability:** Do processes reflect regional infrastructure needs and capacity differences? Do they support resources for smaller organisations/business/community organisations to navigate the approval process?
6. **Monitoring and Continuous Improvement:** Are there systems in place for regular evaluation including independent third party evaluations and process optimisation?

Data and evidence sources

The assessment process relies on diverse, credible sources to ensure the integrity of the assessment.

Government Documents

- National and state-level planning legislation
- Climate change and infrastructure policy papers
- Regulatory guidelines for environmental approvals e.g. the EPBC Act.
- Reports from regulatory agencies (e.g. Climate Change Authority, Infrastructure Australia, Clean Energy Regulator, Australian Energy Infrastructure Commissioner)
- Budget allocations for approval process improvements

Industry and Independent Reports

- Research from think tanks (e.g. Beyond Zero Emissions)

- Industry submissions to government consultations
- International best practice comparisons

Project and Community Feedback

- Case studies from approved and delayed renewable energy projects
- Public submissions from community consultations
- Stakeholder feedback on approval process experiences

International Benchmarks

- Approvals process analysis from OECD and IEA reports
- Comparisons with leading international jurisdictions (e.g. UK, Germany, Canada)

Scoring Process

1. **Preliminary Analysis:** Review government policies and regulatory documents on approval processes.
2. **Indicator Evaluation:** Assess efficiency, regulatory clarity, environmental integration, and adaptability to zero-emissions technology.
3. **Stakeholder Insights:** Engage with project developers, policymakers, and industry experts to gather feedback on approval processes.
4. **Case Study Analysis:** Examine approval timelines, regulatory challenges, and project outcomes from recent zero-emissions projects.
5. **Scoring Aggregation:** Assign a score based on performance across key criteria.
6. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Process Efficiency	0.2	Approval processes are inefficient, unpredictable, and lengthy	Some structured approval processes exist but remain slow and complex	Timelines are lengthy, but there are emerging efforts to improve predictability	Moderate efficiency with streamlined pathways for common renewable projects	Highly efficient, transparent, and expedited approval pathways for zero-emissions projects
Regulatory Clarity	0.2	Unclear, inconsistent, and overlapping regulatory requirements across jurisdictions	Some standardisation exists, but guidance remains vague or outdated	Increasing clarity with improvements in cross-jurisdictional coordination	Well-defined, transparent, and easy-to-navigate regulatory framework	Highly consistent, transparent, and well-coordinated regulatory system with national harmonisation
Environmental Integration	0.15	Environmental considerations are minimal or weakly enforced	Some environmental safeguards exist, but they lack comprehensive enforcement	Moderate consideration of environmental impacts, with emerging criteria to enhance protections	Strong integration of environmental and emissions criteria in approval processes	Fully embedded environmental and climate impact criteria with science-based decision-making
Support for Innovation	0.15	Approval processes do not accommodate emerging zero-emissions technologies	Some flexibility for innovation, but pathways remain unclear or slow	Emerging frameworks for innovative technologies, but challenges remain	Defined pathways for emerging technologies with improved flexibility	Adaptive and innovation-friendly processes, ensuring rapid approval for emerging clean technologies
Regional Adaptability	0.15	Processes do not consider regional infrastructure needs or resource limitations	Limited support for regional variations, small businesses, and community organisations	Some regional adaptability with occasional dedicated resources	Approval processes include structured regional pathways and support programs	Fully tailored regional approval processes ensuring local needs, small-business accessibility, and just transition support
Monitoring & Continuous Improvement	0.15	No tracking or assessment mechanisms to improve the approvals process	Some periodic reviews, but no transparent accountability mechanisms	Increasing monitoring with internal evaluations, but no independent oversight	Strong tracking mechanisms with independent audits and regular policy adjustments	Highly transparent, independent evaluation processes with ongoing optimisation and stakeholder feedback integration

Scoring description

Score	Description	Key Characteristics
1	Minimal Alignment (Severe Deficiency)	- Approval processes are inefficient and inconsistent, with limited prioritisation of zero-emissions projects. - Environmental considerations are minimal.
2	Basic Alignment (Moderate Deficiency)	- Basic provisions for zero-emissions projects exist but approval timelines remain lengthy and guidance lacks specificity.
3	Moderate Alignment (Developing)	- Basic provisions for zero-emissions projects exist, but timelines are lengthy, and environmental criteria are insufficiently detailed (they do not identify and commit to measures to minimise environmental harm and maximise environmental benefits).
4	Strong Alignment (Effective)	- Processes demonstrate moderate efficiency with streamlined pathways for common renewable projects. - Challenges remain for innovative technologies.
5	Exemplary Alignment (Leading Practice)	- Processes are highly efficient, transparent, and adaptive, fully integrating environmental and emissions considerations in decision-making.

3.1.4 Investment Framework

Purpose of assessment: To evaluate the level of support provided by the federal government's investment framework for zero-emissions initiatives. The assessment considers funding availability, strategic alignment with net-zero targets, incentivisation mechanisms, and accessibility of investment streams across sectors.

Assessment framework

Investment frameworks are assessed based on six core dimensions:

1. **Funding Availability:** Are sufficient, diverse funding streams available for various zero-emissions projects?
2. **Strategic Alignment:** Does the framework align with national, regional, and sectoral transition goals?
3. **Incentivisation Mechanisms:** Are there incentives to attract private sector investment and encourage industry transition?
4. **Accessibility and Equity:** Is funding accessible across regions, industries, and innovation stages? Is there sufficient support for SMEs to access these funds?
5. **Monitoring and Adaptability:** Are funding programs monitored, evaluated, and adapted to evolving needs?
6. **Public-Private Collaboration:** Does the framework support partnerships between government, industry, and/or research bodies?

Data and evidence sources

Data will be collected from government reports, budget papers, policy documents, and authoritative reviews (e.g. Climate Change Authority's *Sector Pathways Review*). Sources include the Australian Renewable Energy Agency (ARENA), the Clean Energy Finance Corporation (CEFC), and the Department of Industry, Science and Resources. Data collection includes:

- Federal budget allocations for net-zero initiatives.
- Reports from government agencies overseeing clean energy investments.
- Stakeholder interviews with policymakers, industry representatives, and researchers.
- Cross-referencing national targets with regional investment patterns. The assessment draws on diverse, credible sources to ensure objectivity and comprehensiveness.

Scoring Process

1. **Preliminary Analysis:** Collect and review relevant policy documents and budget papers.
2. **Indicator Evaluation:** Score each indicator based on available data and observations.
3. **Stakeholder Insights:** Gather expert opinions to validate findings.
4. **Scoring Aggregation:** Calculate the average score across indicators.

5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Funding Availability	0.2	Limited or fragmented funding for zero-emissions initiatives	Basic, ad-hoc funding for selected technologies	Structured funding for core renewable energy projects	Robust, multi-stream funding across sectors	Comprehensive funding for diverse zero-emissions technologies
Strategic Alignment	0.2	No strategic alignment with net-zero goals	Partial alignment with regional needs	Emerging alignment with sectoral decarbonisation goals	Strong alignment with net-zero transition needs	Full integration with national and regional strategies
Incentivisation Mechanisms	0.15	No industry-specific incentives	Limited incentives, with bureaucratic challenges	Some incentives available; accessibility varies	Clear and accessible incentives for industry	Innovative, well-integrated incentives that promote public-private collaboration
Accessibility and Equity	0.15	Funding inaccessible to SMEs or regional projects	Some access, but barriers exist for smaller businesses	Funding available but not consistently equitable	Strong funding access across industry sizes and regions	Fully inclusive funding with tailored SME and regional support
Monitoring and Adaptability	0.15	No monitoring or adaptation mechanisms	Basic oversight, but no clear pathways for policy refinement	Some review mechanisms, but lacks structured adaptation	Regular evaluation with adjustments based on impact	Dynamic, evidence-based funding evolution with continuous policy refinement
Public-Private Collaboration	0.15	No structured partnerships between government, industry, and research	Some collaboration, but weak industry engagement	Moderate engagement with growing collaboration efforts	Strong industry-government partnerships for decarbonisation	Fully integrated investment ecosystem fostering innovation and cross-sector cooperation

Scoring description

Score	Description	Key Characteristics
1	Minimal Investment Support (Severe Deficiency)	- Funding: Limited or fragmented funding for zero-emissions initiatives. - Alignment: No strategic alignment with net-zero goals. - Incentivisation: No industry-specific incentives.
2	Basic Investment Support (Moderate Deficiency)	- Funding: Basic, ad-hoc funding for selected technologies. - Alignment: Partial alignment with regional needs. - Incentivisation: Limited incentives, with bureaucratic challenges.
3	Moderate Investment Support (Developing)	- Funding: Structured funding for core renewable energy projects. - Alignment: Emerging alignment with sectoral decarbonisation goals. - Incentivisation: Some incentives available; accessibility varies.
4	Strong Investment Support (Effective)	- Funding: Robust, multi-stream funding across sectors. - Alignment: Strong alignment with net-zero transition needs. - Incentivisation: Clear and accessible incentives for industry.
5	Exemplary Investment Support (Leading Practice)	- Funding: Comprehensive funding for diverse zero-emissions technologies. - Alignment: Full integration with national and regional strategies. - Incentivisation: Innovative, well-integrated incentives that promote public-private collaboration.

3.1.5 Research and Development Framework

Purpose of assessment: To evaluate the level of support provided by the Australian Government's research and development (R&D) framework for clean technologies and industrial decarbonisation initiatives. The assessment considers funding levels, strategic alignment with net-zero goals, incentivisation mechanisms, and collaboration with research institutions and industry.

Assessment framework

R&D policies are assessed based on six core dimensions:

1. **Funding Availability:** Are sufficient, diverse funding streams available for research targeting clean technologies? Are the universities/ Cooperative Research Centres sufficiently funded to take the tech for conceptualisation to lab tests?
2. **Strategic Alignment:** Does the framework align with national, regional, and sectoral transition goals?
3. **Collaboration and Partnerships:** Are there strong partnerships between government, industry, and research institutions?
4. **Commercialisation and Market Readiness:** Is there support for translating research into market-ready solutions?
5. **Monitoring and Adaptability:** Are R&D programs reviewed and adapted based on technological advancements and climate goals?
6. **International Engagement:** Does the policy foster international collaboration and position the country as a global leader in zero-emissions research?

Data and evidence sources

Data sources include government reports, budget papers, policy documents, industry reviews, and authoritative analyses (e.g., Climate Change Authority's *Sector Pathways Review*). Data collection includes:

- Federal budget allocations for zero-emissions R&D initiatives.
- Reports from ARENA, CEFC, and the Department of Industry, Science and Resources, universities, CSIRO
- Stakeholder interviews with industry experts, policymakers, and researchers.
- Collaboration levels between national and international research bodies.

Scoring process

1. **Preliminary Analysis:** Review government policy documents and funding programs.
2. **Indicator Evaluation:** Assess funding levels, strategic alignment, and collaborations.
3. **Stakeholder Insights:** Gather input from experts in government, industry, and academia.
4. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.

5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Funding Availability	0.2	Limited funding for zero-emissions R&D, minimal support for universities and CRCs	Some funding available, but lacks a comprehensive strategy	Established funding streams for core zero-emissions technologies	Robust, targeted funding for critical sectors and emerging technologies	Extensive, diversified funding covering all facets of zero-emissions R&D
Strategic Alignment	0.2	No alignment with net-zero or decarbonisation strategies	Some alignment, but gaps remain in key sectors	Emerging alignment with national and sectoral decarbonisation goals	Strong alignment with sector-specific and regional transition plans	Fully integrated with national and global net-zero transition goals
Collaboration & Partnerships	0.15	Minimal engagement with industry and research institutions	Some partnerships exist, but collaboration is limited	Increasing partnerships with universities, industry, and CRCs	Deep, structured collaborations across sectors	Strong, well-integrated research ecosystem fostering national and international partnerships
Commercialisation & Market Readiness	0.15	No structured pathways for commercialising clean technologies	Basic commercialisation support, but lacking market incentives	Some commercialisation pathways exist, but challenges remain	Well-developed funding and support for scaling emerging technologies	Highly effective commercialisation pathways, accelerating clean tech adoption and industrial applications
Monitoring & Adaptability	0.15	No tracking or evaluation mechanisms for R&D initiatives	Basic review structures exist, but no structured adaptability	Some monitoring mechanisms, but limited policy refinement	Well-established evaluation and feedback systems	Dynamic, evidence-based policy refinements ensuring continuous R&D optimisation
International Engagement	0.15	No engagement in global research efforts or collaborations	Limited participation in international partnerships	Growing engagement in global cleantech research networks	Strong international partnerships, fostering knowledge exchange	Positioned as a global leader in zero-emissions research and innovation

Scoring description

Score	Description	Key Characteristics
1	Minimal Support (Severe Deficiency)	- Limited funding for zero-emissions R&D, minimal engagement with industry and research institutions, lack of strategic direction for sustainable innovation.
2	Basic Support (Moderate Deficiency)	- Some funding available but lacking a comprehensive strategy, modest collaboration with research institutions, fragmented support for early-stage low-emission technologies.
3	Moderate Support (Developing)	- Established funding streams for core zero-emissions technologies, increasing prioritisation of climate-related R&D, emerging partnerships with universities, research bodies, and industry.
4	Strong Support (Effective)	- Robust, targeted funding mechanisms for critical sectors, comprehensive R&D strategies aligned with net-zero goals, deep collaborations with industry, academia, and international stakeholders.
5	Exemplary Support (Leading Practice)	- Extensive, diversified funding covering all facets of zero-emissions R&D, strong alignment with national net-zero transition plans, highly collaborative research ecosystem with rapid commercialisation pathways.

3.1.6 Government Procurement Support

Purpose of assessment: To evaluate the effectiveness of federal government procurement policies in advancing zero-emissions objectives through a structured, transparent, and evidence-based framework. The assessment focuses on policy clarity, supplier incentives, implementation consistency, and contribution to sustainable market development.

Assessment framework

Procurement policies are assessed based on six core dimensions:

1. **Policy Clarity and Requirements:** Do procurement policies include clear, enforceable guidelines for zero-emissions products and services?
2. **Supplier Incentives:** Are financial or preferential incentives provided to suppliers using sustainable practices and low-emission materials? Is there support (funding, training, equipment, staff) for local businesses and First Nations businesses to develop their capacity so they can take up procurement opportunities?
3. **Implementation Consistency:** Are sustainability requirements applied uniformly across government departments and contracts?
4. **Monitoring and Compliance:** Is there a system to track supplier adherence to zero-emissions procurement policies?
5. **Demand-Side Market Development:** Does procurement create stable demand for zero-emissions goods and services?
6. **Innovation and Collaboration:** Do policies encourage new technology adoption and collaboration with industry for improved sustainability practices?

Data and evidence sources

Data sources include government procurement guidelines, policy documents, budget allocations, and reports from oversight bodies. Data collection includes:

- Federal procurement policies and sustainability mandates. Includes legislative documents related to green procurement (e.g. Future Made in Australia Bill, Environmentally Sustainable Procurement Policy)
- Major government tenders and contract requirements, for example Infrastructure Australia, guidelines for major construction investment including the Capacity Investment Scheme and Rewiring the Nation.
- Studies from think tanks, reports on sustainable procurement best practices
- Global comparisons from OECD, UN, and other international bodies
- Supply chain assessments on procurement processes and barriers faced by zero-emission suppliers (e.g. BZE's Cleantech Supply Chains briefing papers and *Make it Here: Building Australia's Clean Industrial Future* report.)
- Interviews with industry stakeholders and procurement officers.

- Supplier compliance and monitoring mechanisms.
- Global best practices in green procurement incentives to highlight opportunities.

Scoring process

1. **Preliminary Analysis:** Review government procurement policies and tender requirements.
2. **Indicator Evaluation:** Assess the extent of sustainability criteria and supplier incentives.
3. **Stakeholder Insights:** Gather input from procurement officers, suppliers, and policy experts.
4. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Policy Clarity and Requirements	0.2	Policies contain vague sustainability mentions, lack enforceable criteria	Some sustainability criteria exist but lack enforcement	Structured guidelines for low-emission products with emerging enforcement	Clear, enforceable sustainability requirements	Comprehensive zero-emissions procurement policy with strict enforcement
Supplier Incentives	0.2	No financial or preferential support for zero-emissions suppliers	Limited incentives exist, but are inconsistently applied	Moderate incentives for sustainable suppliers, but with gaps in funding or access	Strong financial and preferential incentives for sustainable suppliers	Best-practice supplier incentives, including First Nations and SME support programs
Implementation Consistency	0.15	Sustainability requirements are applied inconsistently across departments	Some procurement contracts include sustainability clauses, but enforcement is weak	Increasing consistency in sustainability requirements across departments	Well-integrated procurement policy with cross-government alignment	Fully harmonised and standardised sustainability procurement framework
Monitoring & Compliance	0.15	No tracking of supplier compliance with sustainability policies	Limited monitoring, but no penalties or improvement mechanisms	Moderate compliance tracking, but enforcement gaps remain	Strong compliance enforcement with transparent reporting	Highly structured compliance system with independent audits and full transparency
Demand-Side Market Development	0.15	Procurement does not create meaningful demand for zero-emissions products	Some market creation, but weak industry engagement	Procurement influences market development for key technologies	Well-structured procurement programs supporting stable market demand	Strong procurement leadership actively driving zero-emissions market expansion
Innovation & Collaboration	0.15	No focus on emerging zero-emissions technologies or industry collaboration	Some engagement with industry, but weak incentives for innovation	Moderate industry partnerships with pilot initiatives in place	Strong collaboration with industry and research for sustainable innovation	Fully integrated procurement policies that foster zero-emissions technology adoption and commercialisation

Scoring description

Score	Description	Key Characteristics
1	Minimal Support (Severe Deficiency)	- Policies contain vague sustainability mentions, lack enforceable criteria, and provide little or no supplier incentives.
2	Basic Support (Moderate Deficiency)	- Some sustainability criteria exist, but their application is inconsistent and offers limited incentives for suppliers.
3	Moderate Support (Developing)	- Structured guidelines for low-emission products, with growing supplier incentives and moderate compliance enforcement.
4	Strong Support (Effective)	- Well-integrated sustainability requirements, significant supplier incentives, and consistent monitoring and enforcement.
5	Exemplary Support (Leading Practice)	- Procurement actively drives zero-emissions innovation, with strong supplier partnerships and comprehensive policy enforcement.

3.2 State policy

3.2.1 Emissions Reduction Commitment

Purpose of assessment: To evaluate the relevant state government's emissions reduction commitment in driving progress towards decarbonisation and alignment with the 1.5°C target and net-zero goals.

75% emissions reduction by 2035 below 2005 levels is the benchmark assessed for industry and for the whole economy. However, targets differ by sector. The assessment focuses on target ambition, policy enforcement, funding allocation, sectoral coverage, and regulatory alignment, considering legislative frameworks, sector-specific policies, economic incentives, and institutional oversight mechanisms.

While the state and federal indicators assess emissions reduction commitments at different levels of government, they share several commonalities in their approach and evaluation criteria.

Assessment framework

The assessment framework evaluates the state's emissions reduction commitment based on six key categories:

1. **Target Ambition:** Measures alignment with science-based pathways, interim targets, and zero emission goals. A target of 75% emissions reduction by 2035 below 2005 levels is used as a benchmark for this analysis for the whole economy.
2. **Policy Enforcement:** Examines the presence of legal mechanisms, accountability structures, and effectiveness of implementation.
3. **Sectoral Coverage:** Assesses the inclusiveness of high-emission industries and the integration of multi-sectoral policies.
4. **Economic Incentives:** Evaluates funding, subsidies, and financial support for emissions reduction initiatives.
5. **Stakeholder Engagement:** Considers collaboration with industry, communities, and policymakers to ensure effective policy implementation.
6. **Progress Monitoring:** Reviews the presence of independent oversight bodies and transparent reporting mechanisms.

Data and evidence sources

Data sources include government policy documents, legislative acts, emissions reduction plans, budget reports, and industry analysis. Data collection includes:

- Climate Action Tracker⁷ and similar international rating agencies
- State emissions reduction targets and their legal enforcement mechanisms.
- Policy alignment with state, national, and international net-zero goals.
- Budget allocations and financial incentives for emissions reduction.
- Sector-specific strategies, including energy, transport, industry, and land use.

- Progress monitoring through independent commissions or reporting bodies.

Scoring process

1. **Preliminary Analysis:** Review state emissions reduction policies and legal commitments.
2. **Indicator Evaluation:** Assess target ambition, enforcement mechanisms, and policy coverage.
3. **Stakeholder Insights:** Engage with government officials, industry representatives, and climate researchers.
4. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Target Ambition	0.2	Weak, outdated, or symbolic targets misaligned with 1.5°C pathway	Partial net-zero alignment with limited ambition	Legally binding targets, broadly aligned with net-zero by 2050	Science-based targets with significant 2030 reduction goals	Exceeds international benchmarks, detailed interim milestones
Policy Enforcement	0.2	No enforceable emissions reduction policies	Some policies exist but with weak enforcement	Policies in place with moderate regulatory oversight	Comprehensive policies with strong accountability	Strong regulatory framework with continuous enforcement and oversight
Sectoral Coverage	0.15	No sector-specific policies or industry coverage	Limited sectoral coverage, with gaps in high-emission industries	Multiple sectors addressed but implementation is inconsistent	Comprehensive, enforceable policies across all sectors	Fully integrated multi-sector approach with strong inter-agency coordination
Economic Incentives	0.15	No or minimal funding and incentives for decarbonisation	Some subsidies and incentives exist but with limited reach	Moderate funding for emissions reduction programs	Significant financial incentives supporting industry transitions	Fully funded incentives driving technological and industrial transformation
Stakeholder Engagement	0.15	No engagement with industry, communities, or experts	Limited or one-time stakeholder engagement	Ongoing engagement with industry and community, but inconsistent	Strong collaboration with industry, communities, and academia	Institutionalised stakeholder engagement ensuring continuous feedback and adaptation
Progress Monitoring	0.15	No tracking or independent oversight mechanisms	Some tracking but lacks transparency or enforcement	Periodic progress reporting with moderate oversight	Strong independent monitoring and accountability frameworks	Best-practice reporting mechanisms with third-party verification and adaptive policies

Scoring description

Score	Description	Key Characteristics
1: Minimal Commitment (Severe Deficiency)	Limited or symbolic emissions reduction targets, lacking ambition or enforceability.	- Targets are outdated or unambitious, with symbolic policies lacking enforcement or measurable outcomes. - Few or no sector-specific policies or funding allocations. - Minimal regulatory enforcement and accountability mechanisms.
2: Basic Commitment (Moderate Deficiency)	Short-term targets and policies exist but lack a comprehensive approach.	- Targets address some emissions sources but lack full net-zero alignment. - Limited integration of sectoral strategies and moderate resource allocation. - Some incentives for emissions reduction but inconsistently applied.
3: Moderate Commitment (Developing)	Defined emissions targets with growing sectoral coverage and implementation mechanisms.	- Legally binding emissions targets, generally aligning with net-zero by 2050 - Policies addressing multiple sectors with moderate financial support. - Industry and community partnerships for emissions reduction but uneven implementation.
4: Strong Commitment (Effective)	Coordinated and ambitious emissions reduction targets with robust policy implementation.	- Ambitious, science-based targets aligning with significant reductions by 2030 and 2035 and net-zero by 2050. - Comprehensive, enforceable policies and strong regulatory frameworks. - Well-funded incentives for low emission technologies and sectoral transitions.
5: Exemplary Commitment (Leading Practice)	The State demonstrates leadership in emissions reduction, fully aligned with a pathway to contain global warming to safe levels below 1.5°C.	- Targets exceed international benchmarks, with detailed interim milestones for 2030 and 2035 and net-zero by 2050. - Integrated policies with strong regulatory accountability and clean technology incentives. - Continuous policy review, stakeholder collaboration, and substantial investment in zero-emissions innovation.

3.2.2 Skills and Training

Purpose of assessment: To evaluate the effectiveness of the relevant state government's skills and training policy in facilitating the workforce transition to a zero-emissions economy. The assessment focuses on funding, alignment with industry needs, workforce transition strategies, and long-term planning.

Like the previous indicator, both state and federal indicators assess skills and training policies at different levels of government and share several commonalities in their approach and evaluation criteria. The key distinction here is that this assessment is conducted from a state-based perspective.

Assessment framework

State skills and training policies are assessed based on six core dimensions:

1. **Policy Alignment and Strategy:** Is there a clear, structured plan for training the workforce in zero-emissions sectors, aligned with state decarbonisation goals?
2. **Funding and Resource Allocation:** Are there dedicated financial resources and incentives to support zero-emissions skills training, and workforce development and facilities?
3. **Industry and Educational Collaboration:** Is there active engagement between government, industry, and educational institutions to develop sector-specific training?
4. **Workforce Planning and Strategy:** Is there a clear, strategic workforce transition plan that aligns with national and regional energy transition goals?
5. **Implementation and Accessibility:** Are programs effectively delivered and accessible to affected sectors and communities?
6. **Outcomes and Continuous Improvement:** Are there mechanisms to track progress, refine strategies and adapt to emerging challenges?

Data and evidence sources

Data sources include state government policy documents, budget allocations, industry reports, and stakeholder consultations. Data collection includes:

- State government workforce development policies and investment plans.
- Budget allocations for renewable energy training and skills initiatives.
- Engaging with industry stakeholders, training institutions, and regional workforce development agencies.
- Collaboration mechanisms between government, education providers, and industry partners.

Scoring process

1. **Preliminary Analysis:** Review state workforce development policies and training programs.
2. **Indicator Evaluation:** Assess funding levels, strategic planning, and collaboration efforts.

3. **Stakeholder Insights:** Gather input from industry representatives, training institutions, and government bodies.
4. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Policy Commitment	0.2	No targeted programs or workforce transition recognition	Some recognition but lacks integration with transition planning	Recognises workforce skilling as important but lacks full coordination	Strong commitment with structured policy integration	Comprehensive, long-term commitment with strategic workforce alignment
Funding & Resources	0.2	No funding allocated for workforce transition	Limited or one-off funding for training programs	Some funding but lacks scalability or sustainability	Well-funded, long-term workforce programs with strong institutional backing	Sustainable funding model ensuring long-term industry and regional workforce transformation
Industry & Education Collaboration	0.15	No collaboration between government, industry, and education providers	Some engagement but lacks depth and coordination	Moderate collaboration with emerging industry partnerships	Strong industry partnerships embedded in policy implementation	Deeply integrated government, industry, and education partnerships with continuous feedback mechanisms
Workforce Planning & Strategy	0.15	No workforce transition plan or strategic direction	Basic workforce initiatives exist but lack clear long-term planning	Recognises workforce needs but implementation gaps remain	Comprehensive workforce planning with sectoral strategies	Best-practice workforce strategy, aligned with national and regional economic transitions
Implementation & Accessibility	0.15	No access to reskilling/upskilling programs for workers	Limited access, with barriers for affected workers and communities	Some training programs exist but not widely available or tailored	Well-structured, inclusive training programs supporting multiple sectors	Fully accessible, diverse workforce transition programs with national coverage
Outcomes & Continuous Improvement	0.15	No tracking or evaluation of program effectiveness	Limited monitoring with no process for policy improvement	Some tracking mechanisms but lacks transparent accountability	Well-developed tracking systems with periodic review and improvements	Leading model for adaptive policy, with iterative improvements based on real-world outcomes

Scoring description

Score	Description	Key Characteristics
1: Minimal Support (Severe Deficiency)	The policy has limited provisions for zero-emissions skills, with few initiatives and no clear strategic focus.	- Isolated training initiatives lacking coordination with industry and education providers. - Minimal funding for renewables and low-emission technologies. - Few efforts to reskill workers from carbon-intensive sectors. - No investment in training facilities.
2: Basic Support (Moderate Deficiency)	The policy provides basic support but lacks depth, comprehensive planning, or sustained funding.	- General training programs for clean industries with limited coverage. - Some collaboration with institutions, but inconsistent application. - Limited funding for worker reskilling, particularly in regional areas. - No or low investment in training facilities.
3: Moderate Support (Developing)	The policy demonstrates growing support with aligned programs and increased funding.	- Developed training initiatives for renewable energy, energy efficiency, and green industries. - Moderate collaboration with education and industry partners. - Available funding for workforce reskilling and upskilling. - Low to moderate investment in training facilities.
4: Strong Support (Effective)	The policy provides well-coordinated support, strong funding, and structured partnerships.	- Comprehensive, sector-specific training programs in renewable energy, green construction, and sustainability. - Strong partnerships with institutions, local councils, and industries. - Well-funded reskilling programs and training facilities in all regions.
5: Exemplary Support (Leading Practice)	The policy is highly effective and transformative, fully supporting the zero-emissions workforce transition.	- Extensive funding and accessible training for all key zero-emissions sectors. - Integrated, holistic workforce strategy with ongoing industry collaboration. - Continuous policy evaluation and adaptability to workforce demands. - Strong investment in reskilling for carbon-intensive industries. - Active skills and training communications plan to attract uptake.

3.2.3 Approvals Process

Purpose of assessment: To evaluate the relevant state government's approvals processes for renewable energy and zero-emissions projects in supporting emissions reduction objectives. The assessment focuses on the efficiency, transparency, and consistency of approval mechanisms, as well as the prioritisation of decarbonisation goals.

This assessment shares several commonalities in its approach and evaluation criteria with the federal indicator counterpart. The key distinction is that this assessment is conducted from a state-based perspective.

Assessment framework

The assessment framework evaluates the state's approvals processes based on the following categories:

1. **Efficiency:** Measures approval timelines and procedural streamlining.
2. **Transparency:** Assesses clarity and accessibility of approval processes.
3. **Sectoral Consistency:** Evaluates the uniform application of emissions reduction criteria across industries.
4. **Flexibility for Innovation:** Considers adaptability for emerging zero-emissions technologies.
5. **Inter-Agency Coordination:** Reviews collaboration between regulatory bodies to facilitate timely approvals.

Data and evidence sources

Data sources include government policy documents, environmental approvals frameworks, project case studies, and stakeholder consultations. Data collection includes:

- State approval policies for renewable energy projects and emissions reduction initiatives.
- Approval timelines for various project types.
- Engagement with government agencies, industry representatives, and independent advisory panels.
- Coordination evidence between agencies and the integration of emissions reduction criteria.

Scoring process

1. **Preliminary Analysis:** Review state approval frameworks and policies.
2. **Indicator Evaluation:** Assess efficiency, transparency, and emissions reduction integration.
3. **Stakeholder Insights:** Gather feedback from industry representatives, regulators, and experts.
4. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.

5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Efficiency	0.3	Lengthy and inconsistent approval timelines	Some standardisation but approvals remain slow	Predictable and transparent approval timelines	Expedited approvals for renewables	Fully streamlined, fast-track approvals for all zero-emissions projects
Transparency	0.25	Opaque processes with limited public access	Some public reporting but lacks accessibility	Increasing clarity in approval processes	Fully transparent, well-documented public processes	Open access to all approval decisions with real-time reporting
Sectoral Consistency	0.2	No prioritisation for zero-emissions projects	Some prioritisation but inconsistently applied	Prioritisation mechanisms exist but with gaps	Strong prioritisation of low-carbon projects	Universal prioritisation of decarbonisation across all project types
Flexibility for Innovation	0.15	No special consideration for emerging technologies	Some pathways exist but are difficult to navigate	Moderate adaptability for innovative zero-emissions tech	Dedicated fast-track pathways for clean technology	Fully integrated adaptive approvals for cutting-edge innovations
Inter-Agency Coordination	0.1	No collaboration between regulatory bodies	Some inter-agency engagement, but unstructured	Growing coordination, but efficiency gaps remain	Strong inter-agency alignment reducing project delays	Fully harmonised approvals process across all state regulatory bodies

Scoring description

Score	Description	Key Characteristics
1: Minimal Alignment (Severe Deficiency)	Approval processes include minimal provisions for zero-emissions projects, with limited focus on sustainable development.	- No prioritisation of renewable energy projects. - Lengthy and inconsistent approval timelines. - Minimal integration of emissions impact in assessment criteria.

2: Basic Alignment (Moderate Deficiency)	Basic support exists, but comprehensive prioritisation and streamlined approaches are lacking.	- Standardised timelines exist, but without prioritisation mechanisms for zero-emissions projects. - Moderate emissions reduction considerations in decision-making. - Limited inter-agency coordination, leading to project delays.
3: Moderate Alignment (Developing)	Approval processes moderately align with zero-emissions objectives, with some streamlining for renewable projects.	- Predictable and transparent approval timelines. - Growing focus on emissions reduction within decision-making frameworks. - Established but unevenly applied approval pathways for zero-emissions technologies.
4: Strong Alignment (Effective)	Approval processes prioritise and enable efficient zero-emissions project approvals.	- Expedited approvals for renewable energy and low-emission projects. - Consistent application of emissions reduction criteria across all sectors. - Strong inter-agency coordination to reduce project delays.
5: Exemplary Alignment (Leading Practice)	Approval processes are fully optimised for zero-emissions objectives, supporting innovation and rapid deployment.	- Fully streamlined, fast-track approvals for all zero-emissions projects. - Comprehensive integration of emissions reduction across all regulatory criteria. - High levels of inter-agency and stakeholder collaboration, with continuous process improvements.

3.2.4 Investment Framework

Purpose of assessment: To evaluate the relevant state government's investment framework for the zero-emissions transition in recognising and prioritising investment needs and opportunities through a reputable and future-focused approach. The assessment focuses on funding mechanisms, strategic planning, incentives, and support for clean energy infrastructure, innovative technology, and industry transformation.

This assessment shares several commonalities in its approach and evaluation criteria with the Federal indicator counterpart. The key distinction here is that this assessment is conducted from a state-based perspective.

Assessment framework

The assessment framework evaluates the state's investment approach based on the following categories:

1. **Funding Mechanisms:** Availability, scale, and consistency of investment in zero-emissions projects.
2. **Strategic Planning:** Alignment with net-zero goals and integration across economic sectors.
3. **Industry Support:** Incentives for private sector investment and business transition.
4. **Regional Impact:** Investment in high-emission regions and just transition initiatives.
5. **Implementation and Adaptability:** Efficiency, monitoring, and responsiveness to evolving market needs.

Data and evidence sources

Data sources include government investment policies, budget allocations, financial reports, and stakeholder consultations. Key data collection activities include:

- Reviewing state government investment policies for clean energy, industry decarbonisation, and emerging zero-emissions technologies.
- Analysing funding commitments and financial mechanisms supporting zero-emissions projects.
- Assessing the role of incentives and public-private partnerships in driving investment.
- Evaluating the strategic alignment of investment frameworks with net-zero objectives.

Scoring process

1. **Preliminary Analysis:** Review state investment policies and funding mechanisms.
2. **Indicator Evaluation:** Assess financial support, strategic planning, and incentives.
3. **Stakeholder Insights:** Engage with industry representatives, policymakers, and climate finance experts.
4. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Funding Mechanisms	0.3	Small, fragmented funding pools with low impact	Some renewables funding, but inconsistent	Established funding for clean energy transition	Robust, diversified funding for clean industries	Comprehensive, sustainable funding for all zero-emissions projects
Strategic Planning	0.25	No alignment with state net-zero goals	Limited integration with economic sectors	Emerging alignment with industry transformation needs	Strong integration with net-zero and economic transition plans	Fully aligned with national and regional decarbonisation strategies
Industry Support	0.2	No incentives for private investment	Some incentives exist but lack impact	Moderate industry incentives for clean tech and renewables	Strong industry investment support and transition programs	Best-practice incentives driving large-scale industrial transformation
Regional Impact	0.15	No support for high-emission or transitioning regions	Some regional investment but lacks a just transition focus	Moderate regional investment for emissions reduction	Well-funded regional transition programs and job creation initiatives	Full just transition funding ensuring economic resilience in affected regions
Implementation & Adaptability	0.1	No tracking of investment effectiveness	Some monitoring but lacks adaptability	Moderate evaluation mechanisms in place	Strong monitoring with regular policy adjustments	Best-practice, data-driven investment optimisation ensuring effectiveness

Scoring description

Score	Description	Key Characteristics
1: Minimal Investment Support (Severe Deficiency)	Limited provisions for zero-emissions initiatives, with minimal funding or focus on sustainable industries.	- Small, isolated funding pools with low impact. - Minimal prioritisation of low-emission industries. - Limited alignment with regional and economic needs.
2: Basic Investment Support (Moderate Deficiency)	Basic funding for zero-emissions projects, but lacks depth and strategic coordination.	- Some funding for renewables, but inconsistent. - Moderate alignment with net-zero goals. - Limited incentives for industry transition.
3: Moderate Investment Support (Developing)	Growing investment in zero-emissions projects, with improving resource allocation.	- Established funding for clean energy and sectoral transition. - Supportive policies but uneven implementation. - Some incentives for businesses, with improving accessibility.
4: Strong Investment Support (Effective)	Well-coordinated investment policies prioritising zero-emissions projects.	- Robust funding for renewables, energy storage, and clean industries. - Strong alignment with state economic and climate goals. - Significant incentives supporting industry innovation and regional growth.
5: Exemplary Investment Support (Leading Practice)	Comprehensive, strategic investment framework fully aligned with net-zero ambitions.	- Extensive funding covering all facets of the zero-emissions economy. - Strong support for innovation and emerging technologies. - Sustainable funding mechanisms ensuring long-term transition success.

3.2.5 Research and Development

Purpose of assessment: To assess the relevant state government's research and development (R&D) policies in supporting the zero-emissions transition. The assessment focuses on funding mechanisms, strategic planning, industry collaboration, and support for clean energy technologies, innovation, and commercialisation.

This assessment shares several commonalities in its approach and evaluation criteria with the Federal indicator counterpart. The key distinction is that this assessment is conducted from a state-based perspective.

Assessment framework

The assessment framework evaluates the State's R&D support based on the following categories:

1. **Funding and resources:** availability, scale, and consistency of investment in zero-emissions research.
2. **Strategic Planning:** Alignment with net-zero goals and integration across economic sectors.
3. **Industry and Research Collaboration:** Support for partnerships between universities, research institutions, and industry.
4. **Commercialisation and Innovation:** Pathways for bringing new technologies from research to market.
5. **Policy Adaptability:** Mechanisms for ongoing evaluation and updates based on climate science and technological advancements.

Data and evidence sources

Data sources include government policy documents, budget allocations, industry reports, and stakeholder consultations. Data collection includes:

- State investment policies for clean technology R&D and industry innovation.
- Funding commitments and financial mechanisms supporting zero-emissions research.
- Role of partnerships between government, industry, and research institutions.
- Evaluating the alignment of R&D initiatives with net-zero objectives.

Scoring process

1. **Preliminary Analysis:** Review state R&D policies and funding mechanisms.
2. **Indicator Evaluation:** Assess financial support, strategic planning, and collaboration efforts.
3. **Stakeholder Insights:** Engage with industry leaders, research institutions, and policymakers.
4. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Funding and Resources	0.3	Limited or fragmented funding for R&D	Some funding but lacks consistency	Established funding for core clean energy research	Robust and diversified funding streams	Comprehensive, long-term R&D funding strategy
Strategic Planning	0.25	No alignment with net-zero or innovation goals	Some alignment but lacks coherence	Growing alignment with clean energy innovation needs	Strong integration with economic and sustainability strategies	Fully embedded in national and global decarbonisation frameworks
Industry and Research Collaboration	0.2	No structured collaboration between industry and research	Limited engagement with research institutions	Moderate engagement in research partnerships	Deep partnerships with industry and universities	Best-practice innovation ecosystem with international partnerships
Commercialisation and Innovation	0.15	No pathways for clean technology commercialisation	Basic commercialisation support with limited market pathways	Moderate support for moving research to market	Strong commercialisation funding and market readiness programs	Leading global model for clean tech commercialisation
Policy Adaptability	0.1	No mechanisms for policy review or adaptation	Some evaluation mechanisms but lacks transparency	Moderate review processes in place	Well-established evaluation with industry consultation	Fully dynamic, evidence-based policy refinement processes

Scoring description

Score	Description	Key Characteristics
1: Minimal Support (Severe Deficiency)	R&D policy has minimal focus on zero-emissions projects, with limited resources and low prioritisation.	- Few or isolated programs with little impact. - Minimal funding or industry engagement. - Limited innovation and progress.
2: Basic Support (Moderate Deficiency)	Basic funding exists but lacks coordination, long-term planning, and comprehensive sectoral coverage.	- Some funding and grants for clean energy research. - Limited collaboration with research institutions. - Fragmented support for early-stage development.
3: Moderate Support (Developing)	Growing commitment to zero-emissions R&D with improving resources and partnerships.	- Established funding streams for renewable energy and energy storage research. - Moderate integration with net-zero objectives. - Partnerships with industry and academia.
4: Strong Support (Effective)	Well-coordinated R&D policy prioritising zero-emissions projects with significant investment.	- Robust funding across key sectors such as renewables, electrification, and green manufacturing. - Comprehensive strategy aligning with net-zero goals. - Deep industry-research partnerships supporting innovation.
5: Exemplary Support (Leading Practice)	A leading-edge R&D framework fully supporting the zero-emissions transition.	- Extensive, diversified funding for all aspects of clean technology research. - Strong collaboration with national and international stakeholders. - Ongoing evaluation and policy adaptation ensuring continuous innovation.

3.2.6 Local Procurement Policies

Purpose of assessment: To assess the effectiveness of the relevant state government's local procurement policies in supporting zero-emissions objectives and local suppliers. The assessment focuses on policy integration, incentives, enforcement mechanisms, and alignment with state sustainability goals.

This assessment shares several commonalities in its approach and evaluation criteria with the federal indicator counterpart. The key distinction is that this assessment is conducted from a state-based perspective.

Assessment framework

The assessment framework evaluates the state's procurement policies based on the following categories:

1. **Policy Integration:** Strength of sustainability and emissions reduction requirements in procurement guidelines.
2. **Incentives for Suppliers:** Availability of financial and non-financial support for local and low-emission suppliers.
3. **Enforcement and Transparency:** Mechanisms ensuring consistent application of procurement policies across departments.
4. **Alignment with Net-Zero Goals:** Extent to which procurement policies contribute to regional and state-wide decarbonisation.
5. **Supplier Engagement and Innovation:** Collaboration with suppliers to advance sustainable procurement practices.

Data and evidence sources

Data sources include government procurement frameworks, budget allocations, industry reports, and stakeholder consultations. Data collection includes:

- Reviewing state procurement policies for sustainability and local sourcing requirements.
- Analysing financial mechanisms supporting zero-emissions suppliers.
- Assessing monitoring and enforcement mechanisms ensuring compliance.
- Evaluating alignment with state net-zero goals and regional economic development strategies.

Scoring process

1. **Preliminary Analysis:** Review government policies and workforce development strategies.
2. **Indicator Evaluation:** Assess funding, industry collaboration, and accessibility of training programs.
3. **Stakeholder Insights:** Engage with industry leaders, training institutions, and workforce representatives.

4. **Case Study Analysis:** Examine existing workforce transition initiatives and their effectiveness.
5. **Scoring Aggregation:** Assign a score based on performance across key criteria.
6. **Review:** Conduct expert validation to finalise the assigned score and identify areas for improvement.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Policy Integration	0.25	Basic sustainability guidelines without enforceable low-emission requirements	Established criteria for sustainable products with moderate prioritisation of zero-emissions suppliers	Clear guidelines requiring low-emission, local suppliers in key procurement areas	Clear guidelines requiring low-emission, local suppliers in key procurement areas	Comprehensive mandates prioritising zero-emissions and local suppliers in all procurement decisions
Incentives for Suppliers	0.2	Weak incentives for local suppliers adopting green practices	Incentives available for green suppliers, but not widely accessible	Significant incentives for suppliers adopting sustainable practices	Significant incentives for suppliers adopting sustainable practices	Extensive financial and strategic support for suppliers transitioning to zero-emissions practices
Enforcement and Transparency	0.2	No enforcement mechanisms, policies applied inconsistently across departments	Consistent application across departments, but room for improved enforcement	Strong compliance mechanisms ensuring sustainability standards are met across departments	Very Strong compliance mechanisms ensuring sustainability standards are met across departments	Full enforcement with independent auditing and transparent reporting
Alignment with Net-Zero Goals	0.2	Low prioritisation of zero-emissions procurement	Moderate alignment with state net-zero goals	Strong alignment with regional and state-wide decarbonisation targets	Very Strong alignment with regional and state-wide decarbonisation targets	Policies fully aligned with state net-zero targets and regional economic development objectives
Supplier Engagement and Innovation	0.15	No collaboration or capacity-building programs for suppliers	Moderate collaboration with suppliers to advance sustainable procurement practices	Strong collaboration with suppliers including innovation incentives	Very Strong collaboration with suppliers including innovation incentives	Extensive, ongoing collaboration and capacity building for suppliers to innovate and adopt zero-emissions practices

Scoring description

Score	Description	Key Characteristics
1: Minimal Support (Severe Deficiency)	Limited provisions for zero-emissions and local procurement, with low sustainability emphasis.	- Basic sustainability guidelines without enforceable low-emission requirements. - Weak incentives for local suppliers adopting green practices. - Low prioritisation of zero-emissions procurement.
2: Basic Support (Moderate Deficiency)	Some consideration of sustainability and local suppliers but lacking consistency and depth.	- Partial inclusion of emissions reduction in procurement policies. - Modest prioritisation of local suppliers but with weak enforcement. - Limited financial incentives for suppliers to adopt green practices.
3: Moderate Support (Developing)	Procurement policies moderately support zero-emissions and local procurement objectives.	- Established criteria for sustainable products, with moderate prioritisation of zero-emissions suppliers. - Incentives available for green suppliers, but not widely accessible. - Consistent policy application across departments but room for improved enforcement.
4: Strong Support (Effective)	Well-coordinated policies prioritising zero-emissions procurement with strong regional benefits.	- Clear guidelines requiring low-emission, local suppliers in key procurement areas. - Significant incentives for suppliers adopting sustainable practices. - Strong compliance mechanisms ensuring sustainability standards are met across departments.
5: Exemplary Support (Leading Practice)	Procurement policies fully aligned with zero-emissions and regional development objectives.	- Comprehensive mandates prioritising zero-emissions and local suppliers in all procurement decisions. - Extensive financial and strategic support for suppliers transitioning to zero-emissions practices. - Continuous review and enhancement of procurement policies to support sustainable innovation.

3.3 Local Government Policy

3.3.1 LGA Decarbonisation Plans

Purpose of assessment: To assess the quality and effectiveness of Local Government Area (LGA) decarbonisation plans in driving emissions reduction. The evaluation focuses on targets, sectoral strategies, funding allocation, stakeholder engagement, and progress monitoring.

Assessment framework

LGA decarbonisation plans are assessed based on six core dimensions:

1. **Target Setting and Strategy:** Does the plan include clear, measurable emissions reduction targets with sectoral breakdowns?
2. **Funding and Resource Allocation:** Are there dedicated financial resources, incentives, or external funding streams supporting implementation?
3. **Sector-Specific Actions:** Does the plan address emissions reductions across energy, transport, buildings, industry, and waste sectors?
4. **Stakeholder Engagement and Collaboration:** Is there strong engagement with local businesses, communities, and industry stakeholders?
5. **Monitoring and Accountability:** Are there mechanisms in place for progress tracking, reporting, and adaptive improvements?
6. **Social Equity and Just Transition:** Does the plan consider social impacts, ensuring an inclusive and fair transition for workers and communities?

Data and evidence sources

Data sources include LGA climate action plans, policy documents, budget reports, and stakeholder engagement summaries. Data collection includes:

- Reviewing official LGA decarbonisation strategies and associated policies.
- Analysing budget allocations for emissions reduction initiatives.
- Conducting interviews with council representatives and local stakeholders.
- Assessing monitoring and reporting mechanisms within LGA plans.

Scoring process

1. **Preliminary Analysis:** Review official LGA decarbonisation plans and related policies.
2. **Indicator Evaluation:** Assess emissions reduction targets, sectoral actions, and funding.
3. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
4. **Stakeholder Insights:** Gather input from LGA officials, businesses, and community groups.
5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Target Setting and Strategy	0.3	No clear decarbonisation plans exist. Climate action is ad-hoc, reactive, or not considered a strategic priority by the council.	A plan or strategy document exists, but it lacks clear, measurable, or time-bound emissions reduction targets. Goals are aspirational (e.g., "work towards reducing emissions") rather than quantitative.	The council has defined, publicly stated emissions reduction targets for its own operations. The overall strategy is clear but may lack detailed sectoral pathways or ambitious alignment with a 1.5°C trajectory.	A clear and comprehensive decarbonisation strategy is in place with ambitious, time-bound targets for both council operations and the wider community that are approaching a 1.5°C alignment.	Best practice is demonstrated. The plan includes formally adopted, science-based targets for the entire LGA that are fully aligned with a 1.5°C pathway. A clear, funded roadmap with accountability for delivery is in place.
Funding and Resource Allocation	0.2	Severely limited or no dedicated funding for climate action. Initiatives are unfunded or entirely dependent on securing external grants.	Some fragmented funding is available through various departmental budgets, but there is no dedicated or consistent central fund for climate action.	The council makes adequate and consistent budget allocations for specific, planned climate projects (e.g., in waste or energy).	Sufficient and stable internal funding mechanisms are established (e.g., a dedicated climate fund or levy), complemented by a proactive and successful strategy for securing external grants and partnerships.	Extensive and diverse internal and external resourcing is secured. This includes a multi-year, protected climate action budget and the successful leveraging of private investment to ensure all strategic priorities are fully funded.
Sector-Specific Actions	0.1	Limited or no meaningful action in key emissions sectors. Any actions are isolated to a single area (e.g., recycling only) or are purely symbolic.	Concrete action is underway in some sectors, but other major community emission sources (such as transport or buildings) are not being systematically addressed.	There is evidence of meaningful action across most key sectors (e.g., energy, waste, transport), but the scale and integration of these actions may be limited or still in pilot phases.	Fully developed plans are being implemented across all key sectors, with clear evidence of scaled-up projects (e.g., EV fleet transition, LED streetlight rollout, building retrofit programs).	Comprehensive, integrated, and transformative measures are active and scaled across all sectors. The council is a leader in implementing innovative, high-impact solutions.
Stakeholder Engagement and Collaboration	0.2	Poor or limited engagement. Decisions are made with minimal public consultation, and no established partnerships for climate action exist.	Some collaboration exists, but it is inconsistent or misses key stakeholders such as major local industries, business chambers, or First Nations groups.	The majority of key stakeholders are engaged, and some collaborative projects are occurring. Engagement is becoming more proactive but may not be consistently deep or continuous.	Widespread and proactive stakeholder engagement is standard practice, with established partnerships that result in joint projects and shared outcomes.	Best practice. The council facilitates extensive, wide-scale community and industry engagement that leads to co-designed, collaborative local action and widespread community buy-in.

Monitoring and Accountability	0.1	Limited or no public monitoring. Progress against any stated goals is not tracked or reported, and there is no accountability for delivery.	Some internal reporting on metrics occurs, but it is not regular, public, or tied to specific emissions reduction targets.	The council has clear benchmarks for action (e.g., number of solar panels installed) that can be tracked over time, with some public progress updates.	Clear benchmarks are established and incorporated into regular, public, annual reporting requirements, showing progress against key actions and targets.	Best practice. The council maintains and publishes a comprehensive, LGA-wide emissions inventory and provides annual public reports on progress against science-based climate benchmarks, ensuring full accountability.
Social Equity and Just Transition	0.1	Limited or no consideration is given to the social impacts of the transition or the needs of vulnerable community members.	There is some consideration for social equity, but the approach is fragmented (e.g., a single program for low-income households) and not integrated across the overall strategy.	Some specific social equity and just transition measures are in place, such as skills training programs or support for social housing, but an overarching strategy may be lacking.	Social equity and just transition measures are proactively implemented and well-integrated into all aspects of decarbonisation planning and policy.	The plan is built on a comprehensive and fully implemented social equity framework to ensure "no-one is left behind," with dedicated support for affected industries and workers and inclusive decision-making processes.

Scoring description

Score	Description	Key Characteristics
1	Minimal Plan (Severe Deficiency)	- Plan exists but lacks depth, funding, and clear sector-specific actions. Relies heavily on external funding.
2	Basic Plan (Moderate Deficiency)	- General emissions reduction targets exist with some actions, but there is limited coordination, funding, and monitoring.
3	Moderate Plan (Developing)	- Defined goals and active initiatives in key sectors, moderate funding, and growing stakeholder engagement.
4	Strong Plan (Effective)	- Comprehensive plan with ambitious targets, sectoral strategies, secured funding, and strong engagement.
5	Exemplary Plan (Leading Practice)	- Fully integrated strategy with ambitious targets, strong funding, innovative partnerships, and continuous adaptation.

4. Social Acceptance

Definition: Approval and acceptance granted by communities impacted by the implementation of REIPs and their upstream transmission and renewables projects.

Measure: Community benefits and impacts from the rollout of renewable energy.

Why is meeting this criteria critical? Communities are critical stakeholders for the timely deployment of renewable energy to meet emission reductions and deliver long term economic prosperity.

4.2 Community Benefits and Impacts

4.2.3 Strategic and Nature Positive Land-Use Planning

Purpose of assessment: A just and effective transition requires renewable energy infrastructure to be developed in harmony with biodiversity, agriculture and other land uses. Strategic and nature-positive land-use planning provides a framework to balance these priorities, supporting a holistic and sustainable approach to large-scale infrastructure development while promoting strong social benefit practices.

Assessment framework

Strategic and nature-positive land-use planning is evaluated across six key dimensions:

1. **Data Availability and Quality:** The extent and reliability of data across key land-use types, including:
 - a. Renewable energy infrastructure (generation, storage and transmission)
 - b. Agriculture
 - c. Biodiversity
2. **Regional Strategic Planning:** The degree to which regional planning processes integrate biodiversity, agriculture, industry, renewable energy infrastructure, and community needs.
3. **Biodiversity Prioritisation in Planning:** How well biodiversity priorities are understood, mapped, and incorporated into development and planning processes for renewable infrastructure.
4. **Consensus and Collaboration:** The level of collaboration and consensus between local communities, state and federal governments, industry, and researchers in land-use planning.

Data and evidence sources

Data sources include:

- Regional and state biodiversity and agricultural land use and renewable infrastructure databases
- Academic and Non-government organisation (NGO) land-use planning reports
- State and federal government planning processes and legislation
- Local and regional strategic plans and pilots (e.g. [Bioregional Plan pilots](#) undertaken to support [EPBC regional planning based decision making processes](#) in partnership with state and federal governments)

Scoring process

1. **Preliminary Analysis:** Review regional datasets, planning processes, legislation, and strategic plans.
2. **Indicator Evaluation:** Assess the availability and quality of datasets, the strength of strategic planning, the degree of biodiversity prioritisation, and the extent of collaboration.
3. **Scoring Aggregation:** Calculate the overall assessment score based on weighted indicators.
4. **Stakeholder Insights:** Gather input from academia, environmental NGOs, government, and community groups.
5. **Review:** Conduct expert validation to finalise the assigned score.

Weighting table

Dimension	Weight	Score				
		1	2	3	4	5
Data Availability & Quality – Biodiversity	0.16	National granularity biodiversity value data sets available for public use	State wide, state granularity biodiversity data sets available for public use	Regional and state based data sets of biodiversity data available for public use	Regional and state based data sets of biodiversity data available for public use, with additional information on ecological connectivity and other key requirements for Matters of National, State and regional Environmental Significance	Comprehensive system in place to prioritise Matters of National, State and regional Environmental Significance at the regional level, underpinned by land use mapping with an understanding of what good looks like to maximise protection of MNES etc.
Data Availability & Quality – Agriculture	0.16	National granularity agricultural land use value data sets available for public use	State wide, state granularity agricultural land use data sets available for public use	Regional and state based data sets of agricultural land use data available for public use	Regional and state based data sets of agricultural land use data available for public use, with additional information on how to prioritise this land use based on agricultural and community values	Comprehensive system in place to prioritise prime agricultural land at the regional level, underpinned by land use mapping with an understanding of what good looks like to maximise protection of prime agricultural land.

Data Availability & Quality – Renewable Energy	0.16	No renewable energy infrastructure data available publicly	Point / location based renewable energy infrastructure data available but hard to access	Point / location based renewable energy infrastructure data available and easy to access for desktop analysis	Infrastructure-footprint based renewable energy infrastructure data available and easy to access for desktop analysis	Infrastructure-footprint based renewable energy infrastructure data available and easy to access for desktop analysis. Estimates available around supporting infrastructure such as roads, services, etc.
Regional Strategic Planning	0.2	No system in place to collate and compare datasets across these themes	Pilot systems in place such as pilot bioregional planning process (e.g. EPBC related federal / state partnership pilot Bioregional Plans)	Scaling up of strategic regional plans being implemented across the region.	Integrated planning progressing, with impactful insights and clear directions, however implementation is still somewhat confusing and complicated	An understanding of areas suitable or not suitable for renewable energy development, the buffers required around infrastructure to minimise impacts on biodiversity, along with best practice infrastructure design to minimise impacts on biodiversity and agriculture. Complemented by an understanding of critical energy, supporting and community infrastructure needs.
Biodiversity Prioritisation in Planning	0.16	No state wide biodiversity prioritisation in place	Research community has progressed principles on good biodiversity planning approaches for renewable energy and other infrastructure but implementation is still not clear	Research community has developed state wide prioritisation of high biodiversity / Matters of State and National Environmental Significance however buy in from industry has not been obtained	Research community has progressed principles on good biodiversity planning approaches for renewable energy and other infrastructure and this is enabling input into state planning processes	Clear pathway for best placement of renewables is determined and being applied to prioritise biodiversity and Matters of State and National Environmental Significance
Consensus and Collaboration	0.16	No collaboration or process in place to build consensus on approach	Community based or pilot government processes in place to build consensus on approach	Progressing government processes in place to build consensus on approach	Strong collaboration and buy in across land use sectors with consensus on approach	Strong collaboration and buy in across land use sectors with consensus on approach, implementation enabled, trialed and applied

Scoring description

Score	Description	Key Characteristics
1	Minimal level of planning (Severe Deficiency)	- Agricultural and biodiversity databases only available at the national level, with no publicly available renewable energy database - No system in place to compare databases - No state-wide biodiversity prioritisation in place - No collaboration between government, community, industry and researchers
2	Basic level of planning (Moderate Deficiency)	- State-level agricultural and biodiversity datasets are publicly available - Location-based renewable infrastructure databases exist, but with access limitations - Pilot programs in place for regional planning and community involvement - Prioritisation principle developing in research, but with no implementation path
3	Moderate level of planning (Developing)	- Regional agricultural and biodiversity datasets are publicly available - Location-based renewable infrastructure databases exist and are easily accessible - Strategic planning pilots are scaling up - Prioritisation principles established in research, but with no industry buy in - Progressing government practices to involve communities
4	Strong level of planning (Effective)	- Regional agricultural and biodiversity datasets are publicly available, with transparency over government declarations - Footprint-based renewable infrastructure databases exist and are easily accessible - Integrated planning processes in place and progressing - Strong collaboration across land-use sectors
5	Exemplary level of planning (Leading Practice)	- Regional agricultural and biodiversity datasets are publicly available, with clarity over what good looks like - Footprint-based renewable infrastructure databases exist and are easily accessible, including surrounding infrastructure - Well-established and comprehensive planning processes incorporating land-use planning best practice - Clear pathway for best placement of renewables - Strong collaboration across land-use sectors with consensus on approach

References

1. IPCC. (2021). *Climate Change 2021 The Physical Science Basis*.

https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_SPM_final.pdf

2. Climate Change Authority. (2024). *2024 Issues Paper: Targets, Pathways and Progress* [Issues Paper].

Commonwealth of Australia.

<https://www.climatechangeauthority.gov.au/sites/default/files/documents/2024-04/Issues%20paper%20-%20Targets%2C%20Pathways%20and%20Progress.pdf>

3. Malos, A., Skarbek, A., & Li, M. (2023). How could Australia actually get to net zero? Here's how. *The Conversation*.

<http://theconversation.com/how-could-australia-actually-get-to-net-zero-heres-how-217778>

4. Australian Energy Market Operator (AEMO). (2024). *2024 ISP - Green Energy Exports—Core, 2024 ISP generation and storage outlook* [Excel Spreadsheet].

<https://aemo.com.au/-/media/files/major-publications/isp/2024/supporting-materials/2024-isp-generation-and-storage-outlook.zip?la=en>

5. Australian Energy Regulator. (n.d.). *Renewable Energy Zones*. Retrieved April 16, 2025, from

<https://www.aer.gov.au/about/strategic-initiatives/renewable-energy-zones>

6. Drew, G. (2014). *Renewable Energy Superpower*. Beyond Zero Emissions.

<https://21255462.fs1.hubspotusercontent-na1.net/hubfs/21255462/Reports%20-%20Research/BZE%202015%20Renewable%20Energy%20Superpower.pdf>

7. Climate Action Tracker. (2024, November 13). *Australia*. Climate Action Tracker.

<https://climateactiontracker.org/countries/australia/>

Appendix A – Indicator Version History

1. Enabling Infrastructure	Indicator Version	Date Updated	Content Updated
1.1 Energy System			
1.1.1 Renewable energy generation	Version 2	October 2025	Additional dimension assessed, weighting table added, score description updated, data source updated
1.1.2 Transmission network	Version 2	October 2025	Dimensions updated, weighting table added, score description updated
1.1.3 Distribution Network	Version 2	October 2025	Dimensions updated, weighting table added, score description updated
1.1.4 Energy Storage	Version 2	October 2025	Additional dimension assessed, weighting table added, score description updated, data source updated
1.1.5 Hydrogen Network	Version 2	October 2025	Additional dimension assessed, weighting table added, score description updated
1.1.6 Energy system orchestration	-		
1.2 Transport System			
1.2.1 Port	-		
1.2.2 Rail	-		
1.2.3 Road	-		
1.2.4 Transport	-		
1.3 Water System and Waste Management			
1.3.1 Water supply	-		
1.3.2 Wastewater management	-		
1.3.3 Solid waste & resource recovery	-		
1.4 Social Infrastructure			
1.4.1 Housing	-		
1.4.2 Health	-		
1.4.3 Education	-		

1.5 Communications Infrastructure			
1.5.1 Digital infrastructure	-		

2. Industry Capability & Capacity	Indicator Version	Date Updated	Content Updated
2.1 Industry Decarbonisation			
2.1.1 Industry decarbonisation plans	Version 2	October 2025	Weighting table added, score description updated.
2.2 Workforce Capacity to Deliver			
2.2.1 Existing workforce capacity	-		
2.2.2 Reskilling and upskilling programs	-		
2.3 Innovation & Knowledge Development			
2.3.1 Innovation	-		
2.3.2 Commercialisation	-		
2.3.3 Knowledge generation and sharing	-		
2.4 Collaboration & Industrial Ecosystems			
2.4.1 Industrial symbiosis & material exchange	-		
2.4.2 Coordinated plan for common user infrastructure	-		
2.4.3 Contribution to Australian supply chain resilience	-		
2.4.4 International linkages / coordination			

3. Policy & Governance	Indicator Version	Date Updated	Content Updated
3.1 Federal Policy			
3.1.1 Emissions reduction commitment	Version 2	October 2025	Weighting table added, dimensions updated

3.1.2 Skills and training	Version 2	October 2025	Weighting table added
3.1.3 Approvals processes	Version 2	October 2025	Weighting table added
3.1.4 Investment framework	Version 2	October 2025	Weighting table added
3.1.5 Research and development	Version 2	October 2025	Weighting table added
3.1.6 Government procurement support	Version 2	October 2025	Weighting table added
3.2 State Policy			
3.2.1 Emissions reduction commitment	Version 2	October 2025	Weighting table added
3.2.2 Skills and training	Version 2	October 2025	Weighting table added, dimensions updated
3.2.3 Approvals processes	Version 2	October 2025	Weighting table added
3.2.4 Investment framework	Version 2	October 2025	Weighting table added
3.2.5 Research and development	Version 2	October 2025	Weighting table added
3.2.6 Government procurement support	Version 2	October 2025	Weighting table added
3.3 Local Government Policy			
3.3.1 LGA decarbonisation plans	Version 2	October 2025	Weighting table added
3.3.2 Regional climate adaptation/ resilience plans	-		
3.4 Governance			
3.4.1 Regional coordination	-		
3.4.2 Regional representation of key agencies	-		
3.4.3 Policy alignment and coordination	-		
4. Social Acceptance	Indicator Version	Date Updated	Content Updated
4.1 Engagement & Participation of Communities			
4.1.1 Knowledge sharing and co-design forums	-		

4.1.2 Effective First Nations engagement	-		
4.1.3 Effective community engagement	-		
4.1.4 Public awareness and education	-		
4.2 Community Benefits & Impacts			
4.2.1 First Nations community benefit sharing	-		
4.2.2 Wider community benefit sharing	-		
4.2.3 Strategic & nature positive land-use planning	Version 1	October 2025	Indicator assessment added
4.2.4 Demographic planning			

5. Financing the Transition	Indicator Version	Date Updated	Content Updated
5.1 Investment			
5.1.1 Public finance commitment	-		-
5.1.2 Attracting private investment	-		-
5.2 Risk & Resilience			
5.2.1 Risk and resilience	-		-